



User Manual

Version 2.3

A music tracker for the POKEY sound chip
of the Atari 8-bit computers (XL/XE)

Based on RASTER Music Tracker by Radek Štěrba (R.I.P.), VinsCool and Peter Dell (JAC!)

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About this manual

This manual describes **RITMO**, a music tracker for the POKEY sound chip of the Atari XL/XE 8-bit computers. **RITMO** (*RITMO Is a Tracker for Music On Atari*) is a fork of the RASTER Music Tracker (**RMT**), and runs on Linux, Windows and macOS.

It is written for two kinds of reader: the musician who wants to compose, and the Atari programmer who wants to use the result in a demo or a game. Chapters 1 to 5 introduce the program and walk through a first song; the next ones describe every part of the window, the menus, the dialogs, the import and export formats, MIDI input and the scripting; the appendices are the complete reference of the keys, the fields and the file format.

Conventions

- Keys are written like **CONTROL+S** or **F5**. **CONTROL** is the **Ctrl** key (on macOS the **Control** key, not **Command**).
- A menu item is written **File** → **Open....**
- Hexadecimal numbers have a **\$** in front: **\$3F** is 63. The tracker shows its numbers in hexadecimal: line numbers, instruments, volumes.
- The names of the fields of the screen are written as the program writes them, in capital letters: **VOLUME**, **TABLE LENGTH**.
- The screenshots are taken from the program itself, by the script `doc/manual/make-screenshots.sh`, with the default settings of a first start.

RMT and RITMO in the reference

The reference tables are converted from the documents of the original **RMT**, and still call the program **RMT** in places. **RMT** is also the name of the file format and of the player routines: an *RMT module* and the *RMT drivers* keep their names in **RITMO**.

Where the reference tables come from

The tables of the menu commands, of the note keys, of the scripting commands and of the tracker drivers (appendix A and chapters 15 and 16) are converted from the Markdown documents in the `doc/` folder of the program, and the menu and note key tables of those are written by the program itself (the `dump actions` and `dump notekeys` script commands). They are always the state of the version this manual is built from.

Contents

About this manual	i
1 Introduction	1
1.1 What RITMO is	1
1.2 History	2
1.3 What is different from RMT	2
1.4 Files used by RITMO	2
2 Installing and starting	3
2.1 The packages	3
2.2 Starting the program	3
2.3 Sound	4
2.4 Where the settings are kept	4
3 Concepts	5
3.1 Time	5
3.2 The sound of the POKEY	6
4 The main window	7
4.1 The areas of the screen	7
4.2 The four editing parts	8
4.3 Edit mode and prove mode	10
4.4 The menus and the toolbars	10
4.5 The mouse	10
5 Your first song	12
5.1 A new song	12
5.2 Entering notes	12
5.3 Choosing the instrument	13
5.4 Playing	13
5.5 More tracks and song lines	13
5.6 Saving and exporting	13

6	Editing tracks	14
6.1	The cursor	14
6.2	The length of a track	14
6.3	Playing while editing	14
6.4	Blocks	15
6.4.1	The Effects and Tools dialog	15
6.5	Tracks as objects	15
6.6	Undo	16
7	Instruments	17
7.1	The parts of an instrument	17
7.2	Editing	17
7.3	Instrument files and the Instrument menu	18
7.4	The distortions	18
7.5	The instrument active help	18
8	The song	19
8.1	Song lines	19
8.2	Subsongs	19
8.3	Bookmarks	19
8.4	The info area	20
8.5	Changing the kind of song	20
8.6	Optimizing the size	21
9	Playing and monitoring	22
9.1	Play commands	22
9.2	Channels	22
9.3	Watching the sound	22
9.4	Playing from the keyboard	23
10	The menus	24
10.1	File	24
10.2	Edit	24
10.3	View	25
10.4	Play	25
10.5	Channels	25
10.6	Song	26

10.7 Instrument	26
10.8 Track	27
10.9 Block	27
10.10Pokey	28
10.11Tools	28
10.12Help	28
11 Options and tuning	29
11.1 The Options dialog	29
11.1.1 General	29
11.1.2 Keyboard	30
11.1.3 MIDI	30
11.1.4 Paths	30
11.2 The tuning	31
12 Import and export	32
12.1 Importing	32
12.2 Exporting	33
12.2.1 RMT stripped	34
12.2.2 ASM simple notation	34
12.2.3 SAP, SAP-R and LZSS	35
12.2.4 XEX	35
12.2.5 Relocatable ASM	36
12.2.6 WAV	36
12.3 Exporting from the command line	36
13 MIDI input	37
13.1 Options	37
13.2 The MIDI channels	38
13.3 Scripts	38
14 The Pokey explorer	39
14.1 The keys	39
15 Tracker drivers	41
15.1 Tracker Drivers	41
15.2 Tracker Driver Versions	41
15.2.1 RMT 1.28 Unpatched by Raster	42

15.2.2	RMT 1.28 Unpatched with Tuning	42
15.2.3	RMT 1.25 Patch3 Instrumentarium by Analmux	42
15.2.4	RMT 1.27 Patch6 by Analmux	42
15.2.5	RMT 1.28 Patch8 by Analmux	42
15.2.6	RMT 1.34 Patch16 by VinsCool	43
15.2.7	RMT 1.28 Patch Prince of Persia by VinsCool	43
16	Scripting	44
16.1	Running a script	44
16.2	Syntax	44
16.3	Commands	45
16.4	Export formats and options	46
16.5	Example	47
16.6	Environment variables	47
16.7	Tests	47
16.8	Not available as script commands	48
A	Reference of the commands, keys and fields	49
A.1	Note Keys	55
A.1.1	QWERTY	55
A.1.2	QWERTZ	56
A.1.3	AZERTY	57
A.2	For All Edit Modes	58
A.2.1	Hotkeys	58
A.3	TRACK EDIT	60
A.3.1	Fields	60
A.3.2	Hotkeys	60
A.3.3	Note Column	61
A.3.4	Instrument Number Column	62
A.3.5	Volume Column	62
A.3.6	Speed Column	62
A.3.7	Block Operations	62
A.3.8	Block Data Modifications	63
A.3.9	Mouse control	63
A.3.10	Tracks top area (Shown as TRACK L1 to TRACK R4 above pattern tracks)	63
A.4	INSTRUMENT EDIT	63

A.4.1	Fields	64
A.4.2	Table of Notes/Frequencies	64
A.4.3	Envelope	65
A.4.4	Sound Type (DISTORTION Parameter)	65
A.4.5	Envelope command (COMMAND parameter)	65
A.4.6	Hotkeys	66
A.4.7	Table of notes/frequencies	66
A.4.8	Envelope Parameters	67
A.4.9	Mouse control	67
A.4.10	Volume Envelope Area	67
A.5	INFO EDIT	67
A.5.1	Hotkeys	68
A.5.2	Mouse control	68
A.5.3	MAXTRACKLENGTH, MONO/STEREO, PAL/NTSC	68
A.6	SONG EDIT	68
A.7	PROVE (JAM) MODE	69
B	The RMT module file format	70
B.0.1	Segment Binary Load Header (Mandatory)	70
B.0.2	Module Header Struct	70
B.0.3	Instrument Struct	70
B.0.4	Table of Notes Struct	71
B.0.5	Envelope Struct	71
B.0.6	Track Struct	71
B.0.7	Instruments Table	72
B.0.8	Track Addresses Table	72
B.0.9	Track Sequence Table	73
C	MIDI implementation chart	74
D	Building from the source	75
D.1	Linux	75
D.2	Windows and macOS	75
D.3	Core only and tests	75
D.4	Test hooks	75
E	Credits and licenses	76

E.1 Credits 76

E.2 The fonts 76

E.3 Disclaimer 76

Introduction

1.1 What RITMO is

The Atari XL/XE computers make their sound with the POKEY chip: four independent voices (sixteen when two chips are used, as in many stereo songs), each with a frequency divider, a choice of noise and tone distortions and a volume. A *tracker* is the kind of music editor that was born on the Amiga: the music is a grid of numbers, one column for each voice and one row for each beat, and the notes are typed with the computer keyboard as on a piano.

RITMO is such a tracker for the POKEY. A song made with it can be played on a real Atari: the program produces the same data the player routines of the Atari read, and exports them in several formats (a SAP file for the Atari players, an executable .xex, assembler source, the compact RMT module for your own program). The sound you hear while composing comes from an emulation of the Atari's 6502 processor and of the POKEY chip built into the program, running the same player routine, so it is what the Atari will play.

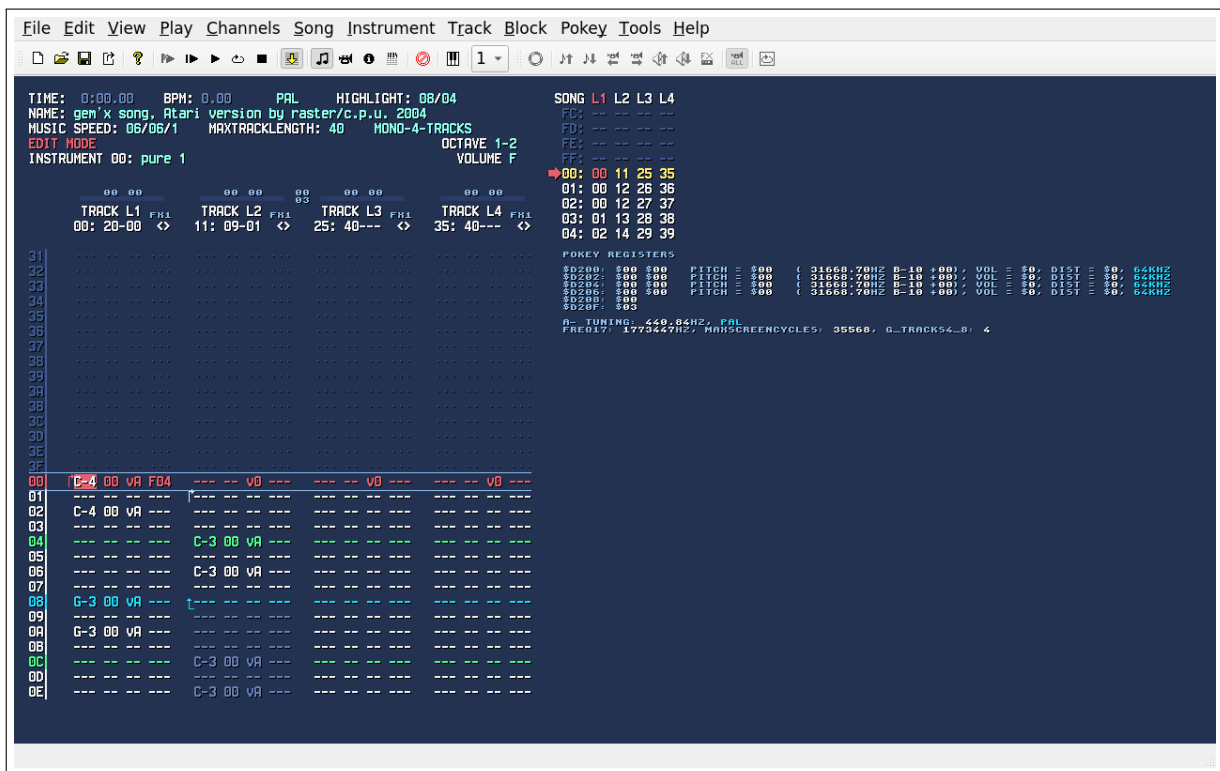


Figure 1.1. The main window of RITMO with a song loaded.

1.2 History

RITMO is a fork of the *RASTER Music Tracker*, **RMT**. The first version, 1.0, was written by Radek Štěrbá (Raster/C.P.U.) in 2002; his last version is 1.28, of 2009. It was a small revolution for the Atari musicians: the first tracker for the POKEY with a graphical envelope editor, and the one nearly everybody used. Radek Štěrbá has since passed away; this manual is, among other things, a way to remember his work. R.I.P.

Vin Samuel (*VinsCool*) continued the program from 2021 to 2024, up to the version 1.34: a new player routine with more tuning tables, a tuning calculator, an automatic high pass filter, the SAP and XEX export and many corrections. Peter Dell (*JAC!*) carries on the original project from 2024 (version 1.35 and the planned version 2) and made it possible for this fork to exist: everything described here is built on his work.

The RITMO fork keeps the code of the tracker, the file formats and the player routines of RMT 1.35, and replaces what tied it to Windows. It is renamed so that it cannot be confused with the original RMT, and so that the Windows-only (MFC) parts of the code could be removed.

1.3 What is different from RMT

- The program is written with Qt: it runs natively on Linux, on Windows and on macOS (Intel and Apple Silicon). The look of the tracker screen is the original one, pixel for pixel.
- The 6502 and the POKEY emulations are built in: the `sa_c6502.dll` and `apokeysnd.dll` libraries of the Windows versions are not needed any more.
- The sound goes out through PortAudio, and the MIDI input comes in through RtMidi.
- The configuration is kept per user in the settings of the system, not in an `.ini` file next to the program. The settings of RMT are taken over at the first start.
- Songs, instruments and tracks are the same files as in RMT 1.3x: you can move your work between the programs. The files keep their extensions, `.rmt` and the others.

1.4 Files used by RITMO

Extension	Content
<code>.rmt</code>	an RMT module: the song, its tracks and its instruments (and, if the song was saved with Save , the names of the song and of the instruments)
<code>.txt</code>	the same song as a text file, which can be read and edited with any editor; a single track is saved as a text file too
<code>.rmw</code>	a song work file: an RMT module with the state of the editing
<code>.rti</code>	a single instrument
<code>.sap</code> , <code>.sapr</code> , <code>.xex</code> , <code>.asm</code> , <code>.wav</code> , <code>.lzss</code>	the exported formats (chapter 12)
<code>.mod</code> , <code>.tmc</code> , <code>.tm8</code>	the formats that can be imported
<code>.rmtscript</code>	a script (chapter 16)

Installing and starting

2.1 The packages

Everything the program needs is inside each package: the tracker drivers and the other files it loads at run time are in the `resources` folder next to the program.

System	Package	How to start it
Linux x86_64 (Debian 11, Ubuntu 20.04 and newer)	Ritmo-Linux-x86_64.AppImage	make it executable (<code>chmod +x</code>) and run it
Windows 64 bit	Ritmo-Windows-x64-Setup.exe, Ritmo-Windows-x64.zip	run the installer (it asks whether to install for all the users or for the current one, and offers a desktop icon and the <code>.rmt</code> file type), or unzip the archive anywhere and run <code>Ritmo.exe</code> (the program is not signed: Windows may ask to confirm)
macOS 12 or later, Apple Silicon or Intel	Ritmo-macOS-arm64.dmg, Ritmo-macOS-x86_64.dmg	drag <code>Ritmo.app</code> to Applications; the first time open it with the right button and Open (the program is not signed)

RITMO can also be built from the source with CMake and Qt (appendix D).

Note

The packages of the versions up to 2.2.1 were called `RMT-*` and the program `rmt` (`Rmt.exe`). From the next releases the packages are `Ritmo-*` and the program `ritmo` (`Ritmo.exe`).

2.2 Starting the program

Start RITMO like any other program. The first start opens the window with an empty song. The program accepts a song on the command line, which is opened at the start:

```
ritmo mysong.rmt
```

With the option `/SCRIPT`: the program runs a script without showing its window (chapter 16):

```
ritmo /SCRIPT:build.rmtscript
```

2.3 Sound

The sound goes out through PortAudio, to the default output device of the system. If the output cannot be opened the program prints a message and goes on without sound, so that songs can still be edited. The *hardware soundbuffer* option of the configuration (section 11.1) and the `RMT_AUDIO_BUFFER_MS` environment variable (the size of the buffer in milliseconds) are for the sound output of the system that needs a different buffer.

2.4 Where the settings are kept

The configuration and the tuning belong to the user, not to the program folder, so that an update or a new installation keeps them:

Linux	<code>~/.config/ritmo-atari.org/ritmo.conf (\$XDG_CONFIG_HOME)</code>
Windows	the registry, <code>HKEY_CURRENT_USER\Software\ritmo-atari.org\ritmo</code>
macOS	<code>~/Library/Preferences/org.ritmo-atari.ritmo.plist</code>

At the first start, if there are no settings of **RITMO** yet, the settings of **RMT** are taken over: those saved by it in the same places with the name `rmt`, or an `rmt.ini` and a `tuning.ini` next to the program.

Concepts

The words of the tracker have a precise meaning in this manual. Outside of it they are often used interchangeably.

Song A piece of music: a sequence of *song lines*. **RITMO** edits one song at a time; a song can start from different song lines, the *subsongs*.

Module

The data structure with the song, its tracks and its instruments, as it is stored in an `.rmt` file (the *module file*).

Track

A logical voice with the notes of a stretch of music, for example eight bars of the bass. It has a number (`$00` to `$FF`), a length (up to 256 lines) and can be used many times in a song.

Song line

A row of the song: it says which track plays on each channel at that moment, or where to *go to*, to loop.

Channel

A physical output to create sound. A mono song has four channels, `L1` to `L4`, one for each POKEY voice; a stereo song has eight, `L1`–`L4` on the left and `R1`–`R4` on the right (two POKEYs).

Instrument

What makes the sound at the different pitches: a volume envelope, a table of notes or frequencies, a choice of distortion, and effects such as vibrato. An instrument is a number (`$00` to `$3F`) and a name.

Pattern

The notes of a track with their attributes (instrument, volume, speed).

Tracker driver

The part of the program that makes the actual sound from the tracker data: a piece of 6502 code that runs on the emulated Atari. Different *versions* of the driver interpret the data in a different way (chapter 15).

Mono / stereo

Mono: the four channels of one POKEY. Stereo: two POKEYs, one for the left and one for the right output.

3.1 Time

The Atari plays the music once for each video frame: 50 times a second on a PAL system and 60 times on NTSC (the **NTSC** option, section 11.1). A *line* of a track lasts `S` frames, where `S` is the *speed* of the song (the **MUSIC SPEED** field). The *engine speed* says how many times the player

routine is called each frame, for the songs that need a finer time. Speeds can change in the middle of a song with the speed column of the tracks.

3.2 The sound of the POKEY

- Each channel has a frequency divider (AUDF), a volume from 0 to 15 and a *distortion* (the AUDC register): pure tone, noise of different kinds, buzzy tones from the polynomial counters.
- The AUDCTL register changes how all the channels work: it chooses the base clock (64 kHz or 15 kHz), joins two channels into a 16-bit divider, adds high pass filters, or clocks a channel with 1.79 MHz. The instruments set these bits (chapter 7).
- The notes of the tracker go from C-1 to C-6: \$00 to \$3D.

The main window

RITMO has one window. Under the menu bar and the toolbars the screen of the tracker is drawn: a fixed picture of the song, with its own font and its own colours, the same as in RMT. The parts of the screen are shown in figure 1.1 (page 1) and described here.

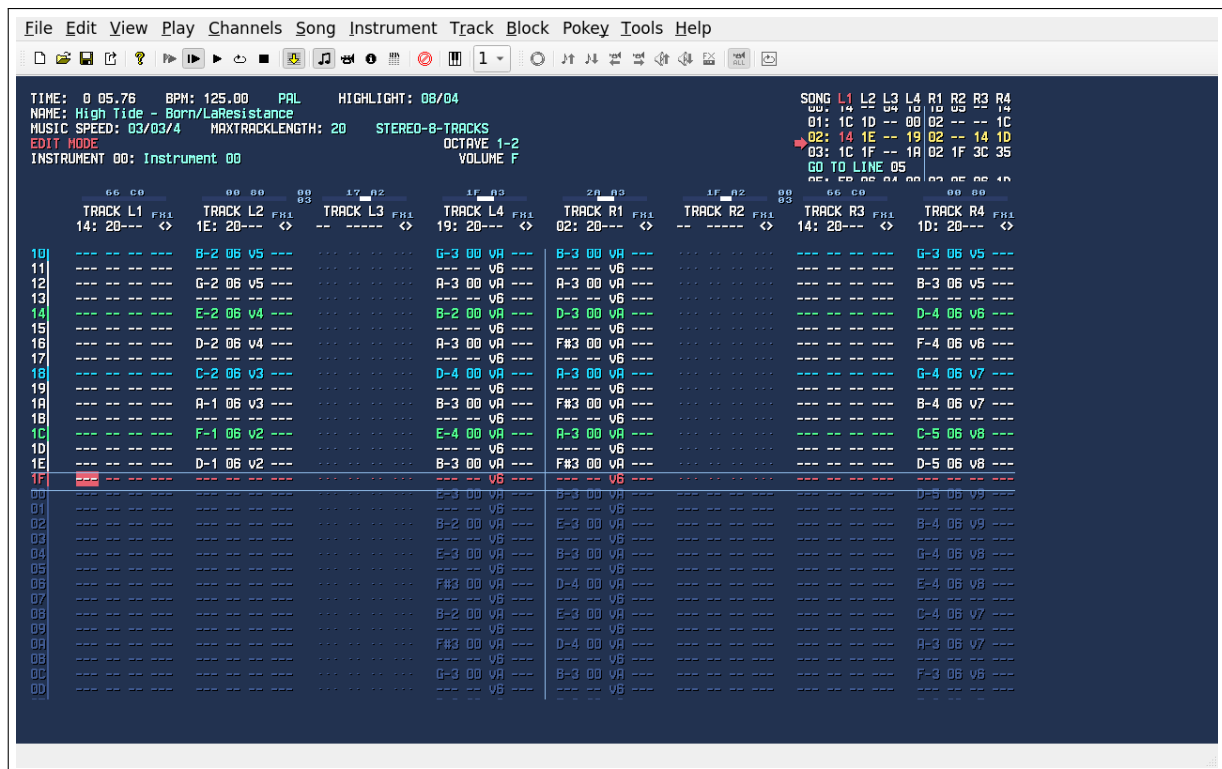


Figure 4.1. A stereo song (8 tracks) playing: the time counter runs, the song line marker follows the music.

4.1 The areas of the screen

Info area (top left)

The state of the program and the data of the song, in a few lines.

- **TIME** and **BPM**: the play time counter (when the counter is enabled in the **View** menu) and the tempo the song speed gives, in beats per minute. **PAL** / **NTSC**: the video system the speed is computed for; the number of frames per second of the screen is shown next to it.
- **HIGHLIGHT**: the two steps of the highlighted lines of the tracks (every 8 lines and every 4 lines in the figure), used to find the beat in the pattern.
- **NAME**: the name of the song and of its author, up to 64 characters.
- **MUSIC SPEED**: AA/MM/S: the current speed, the main speed and the engine speed (chap-

ter 8). **MAXTRACKLENGTH**: the maximal length of the tracks of the song. On the same line the kind of song: **MONO-4-TRACKS** or **STEREO-8-TRACKS**.

- **EDIT MODE** (or **PROVE MODE**, and the other modes of the keyboard), the octave of the new notes (**OCTAVE 1-2**), the active instrument with its name and the volume of the new notes.

Song area (top right)

The sequence of song lines, one row for each line, with a column for each channel (**L1...L4**, and **R1...R4** for a stereo song). Each cell is the number of the track that plays there. The arrow marks the current song line. A line with **GO TO LINE** sends the song to another line: that is how a song loops.

Pokey registers

With **View** → **Pokey Chip Registers** on, the registers of the POKEY (\$D200...), the pitch, volume and distortion of each channel, as the emulated chip has them at that moment.

Tracks area (centre)

The tracks of the current song line side by side, one for each channel, with the lines of the pattern scrolling vertically. The header of a track names its channel (**TRACK L1**) and shows the track number, then its length and the line it loops to (**00: 20-00** is track \$00, \$20 lines long, looping from line \$00). **EMPTY** marks a track with no notes. The cursor is the highlighted cell.

Status bar

The keyboard modes (**CAP** for the caps lock) and, when it is enabled, the debug display.

A line of a track has four columns, drawn like

C-4 00 vA F04

the note, the instrument (hexadecimal), the volume (v and a digit) and the speed command (F and the speed). Empty columns are drawn ---, --.

4.2 The four editing parts

The keyboard acts on one *part* of the screen at a time. The part is chosen with the function keys, and the cursor of the part is the red cell.

Part	Key	What it edits
Tracks (TRACK EDIT)	F2	the notes of the tracks
Instruments (INSTRUMENT EDIT)	F3	the instrument: envelope, table, effects
Song (SONG EDIT)	F4	the sequence of tracks of the song
Info (INFO EDIT)	SHIFT+F4	the name, the speeds, the kind of song

ENTER returns from the song or info part to the tracks or instruments part you came from.

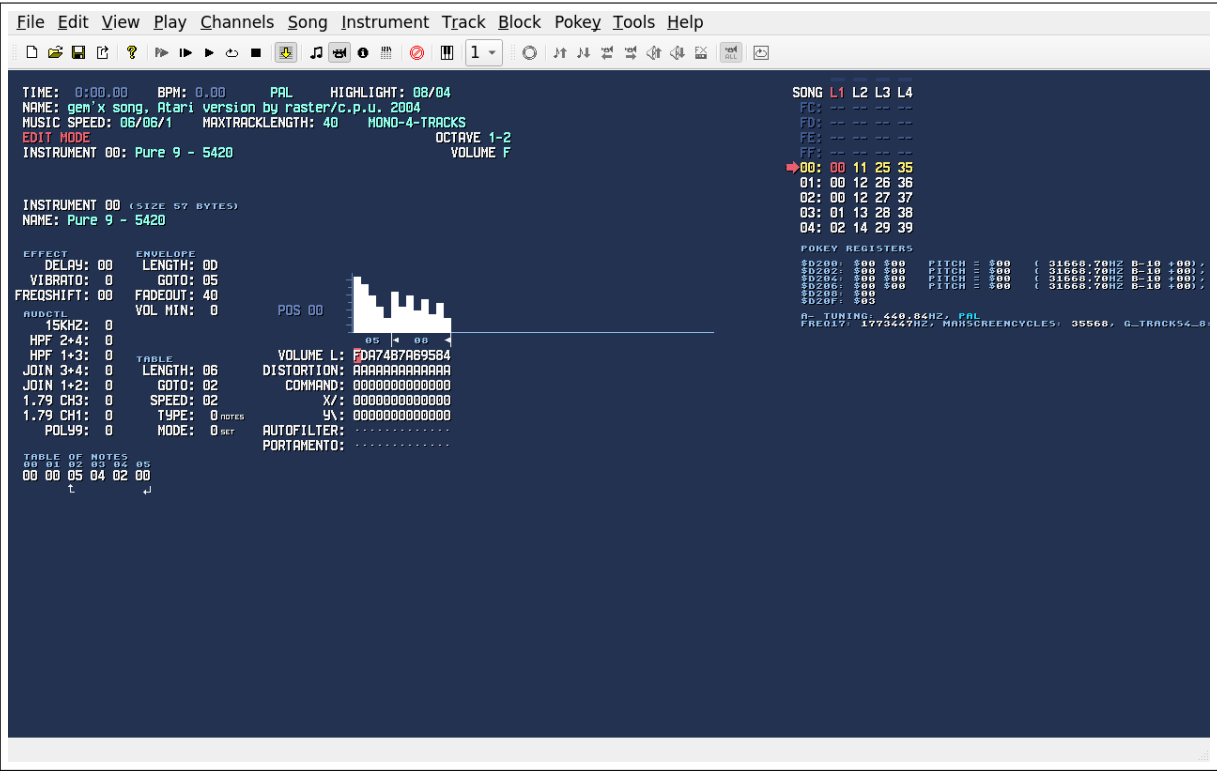


Figure 4.2. The instrument part (INSTRUMENT EDIT) with an instrument loaded.

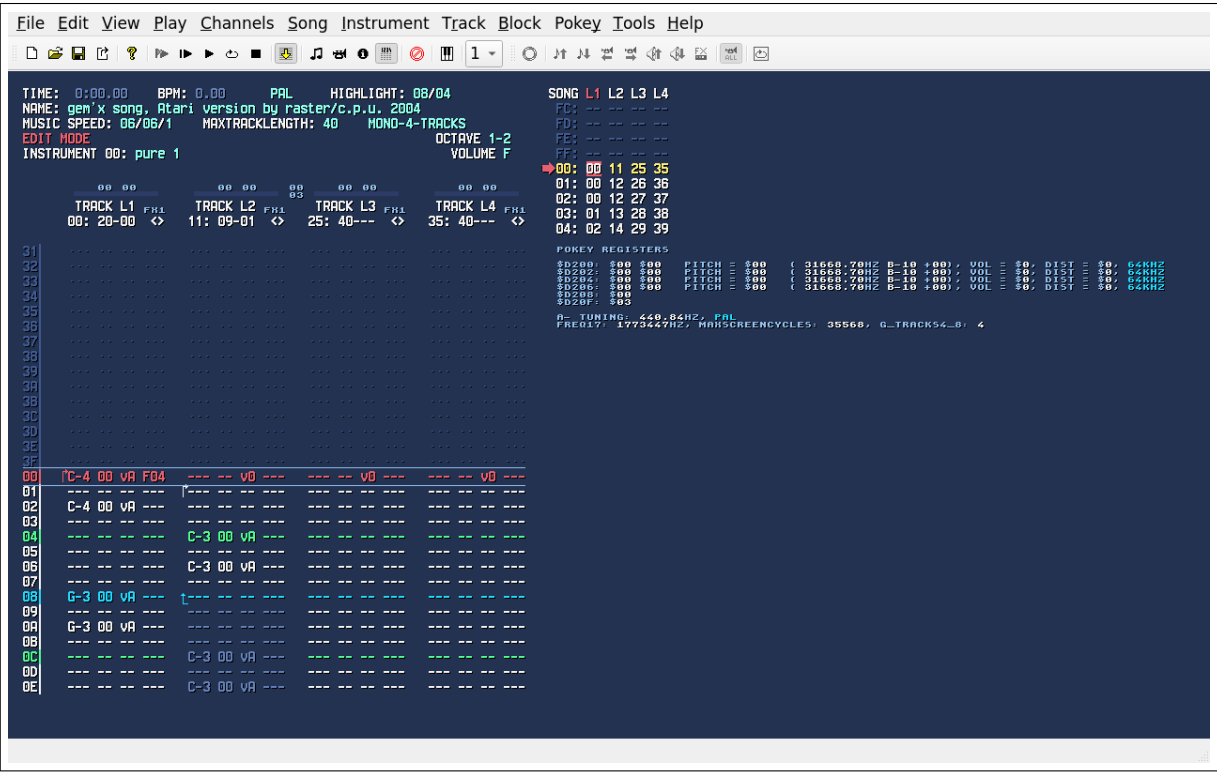


Figure 4.3. The song part: the cursor is in the song area.

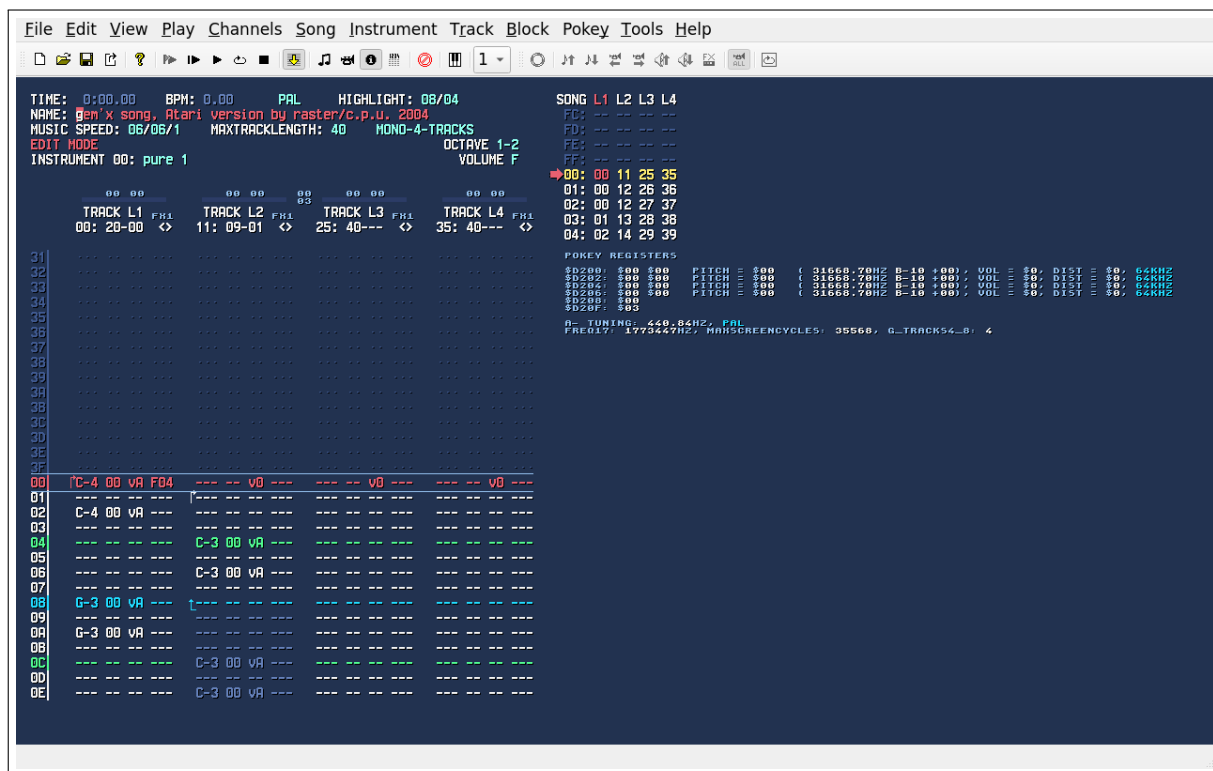


Figure 4.4. The info part: the cursor is on the speed field.

4.3 Edit mode and prove mode

The window is in one of two modes, shown in the info area. In **EDIT MODE** the keys write notes. In **PROVE MODE** (also called *jam* mode) editing is switched off and the note keys only play: you can try an instrument or play along with the song without changing it. **CONTROL+SPACE** (or **Edit** → **Switch Edit Mode**) toggles between them. The instrument and info parts behave as in edit mode; hold **SHIFT** to play notes while there.

4.4 The menus and the toolbars

The menu bar has the menus **File**, **Edit**, **View**, **Play**, **Channels**, **Song**, **Instrument**, **Track**, **Block**, **Pokey**, **Tools** and **Help** (chapter 10). Under them there are two toolbars: the *main toolbar* (new, open, save, export, the play buttons, the edit modes, MIDI on/off...) and the *block toolbar* (transpose, volume, instrument change, play the block...). Both can be shown or hidden in the **View** menu, and the choice is kept in the configuration; the icons follow the interface size. The buttons have a tip with the name of the command and its key.

4.5 The mouse

The program is made for the keyboard, but the mouse helps.

- In the track area: the *left* button on a track header switches the channel on or off, the *right* button plays the channel alone (solo), a second click puts all the channels back.

- In the volume envelope of the instrument: the left button draws the volume, the right one sets it to zero.
- In the info area, on the fields **MAXTRACKLENGTH**, **MONO/STEREO** and **PAL/NTSC**: a click toggles the field or opens the dialog that changes it, then applies the change to the song.
- The wheel over **VOLUME** or **OCTAVE** of the info area changes the volume or the octave of the new notes.

Your first song

This chapter makes a short song from nothing, to see how the parts of the program fit together. Every key used here is explained in the following chapters.

5.1 A new song

Press **CONTROL+N** (**File** → **New**). The dialog of figure 5.1 asks for the *maximal length of the tracks* (how many lines a track can have; 64 is a good start) and for the kind of song: **MONO - 4 TRACKS** or **STEREO - 8 TRACKS**. Choose mono. The current song is discarded, so the dialog warns you.

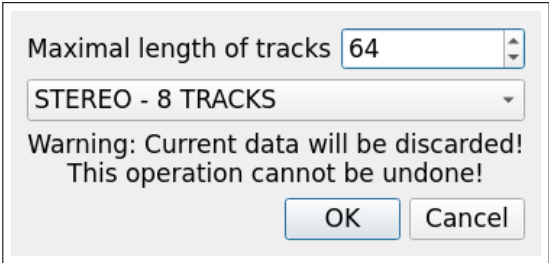


Figure 5.1. The **New** dialog.

5.2 Entering notes

The cursor is in the first column of the first track, the note column. The keyboard is a piano (figure 5.2): the lower row of letters, **ZXC...**, plays from **C-1** and the upper row, **QWE...**, from **C-2**; the keys of the row above each are the sharps (**S, D...** and **2, 3...**). The exact keys depend on the keyboard layout chosen in the options (**QWERTY**, **QWERTZ** or **AZERTY**): the tables of appendix A show every key.

1	2	3	4	5	6	7	8	9	0	-	=
	C#2	D#2		F#2	G#2	A#2		C#3	D#3		F#3
Q	W	E	R	T	Y	U	I	O	P	[	]
C-2	D-2	E-2	F-2	G-2	A-2	B-2	C-3	D-3	E-3	F-3	G-3
A	S	D	F	G	H	J	K	L	;	'	
	C#1	D#1		F#1	G#1	A#1		C#2	D#2		
Z	X	C	V	B	N	M	,	.	/		
C-1	D-1	E-1	F-1	G-1	A-1	B-1	C-2	D-2	E-2		

Figure 5.2. The note keys of the **QWERTY** layout.

Press **Z**: the note **C-1** is written with the active instrument and the volume of the info area, it is played, and the cursor moves down by the number of lines of the *note spacing* (the combo of the

toolbar, default 1). Press `x`, `c...` to write a scale. `SPACE` deletes what is under the cursor, `BACKSPACE` deletes the note.

Tip

`numblock /` and `numblock *` lower and raise the octave of the new notes, `numblock +` and `numblock -` change the spacing of the notes. The volume of the new notes is changed with `CONTROL+numblock +` and `CONTROL+numblock -`.

5.3 Choosing the instrument

The instrument is the sound of the note. The active instrument is shown in the info area (`INSTRUMENT 00: ...`). `SHIFT+LEFT` and `SHIFT+RIGHT` change it. To see what an instrument is, press `F3`: the instrument part of figure 4.2 appears. Chapter 7 explains it; the quickest way to get good sounds is to load instruments: **Instrument** → **Load instrument from file...** and choose one of the `.rti` files in the `instruments` folder of the program.

5.4 Playing

- `F5` plays the song from the start, `F7` from the cursor, `F6` loops the current song line and `ESC` stops.
- `ENTER` plays the note under the cursor.
- `F12` makes the screen follow the playing position.

5.5 More tracks and song lines

The notes of a track belong to a *song line*. To write more music, go to the song area with `F4`: each column is a channel and each number a track. Press `CONTROL+T` to put a new empty track in the current song line and channel, and `F2` to come back and write in it. `CONTROL+D` duplicates the track under the cursor, to change a copy. `CONTROL+G` makes the current song line a `GO TO LINE`, which loops the song.

5.6 Saving and exporting

`CONTROL+S` saves the song as an `.rmt` file. To use it on the Atari, **File** → **Export** makes a SAP file (to play with the Atari players), an executable `.xex` that plays the song when started, assembler source, or a WAV file (chapter 12).

Note

Test the finished song on the Atari or in an Atari emulator, when you can: **RITMO** plays it with an exact emulation of the player routine, but the way the notes sound depends on the *tracker driver version* (chapter 15) and on the video system, PAL or NTSC.

Editing tracks

The tracks are edited in the **TRACK EDIT** part (F2). The complete list of keys of the part is in appendix A; this chapter explains what they do.

6.1 The cursor

The arrow keys, **TAB** and **SHIFT+TAB**, **PAGE UP** and **PAGE DOWN** move the cursor in the tracks. The columns of each track are, from the left: the note, the instrument, the volume and the speed. The key you type goes to the column under the cursor.

Column	What you type
Note (NNN)	a note key; numblock 1–6 change the octave of the note under the cursor; BACKSPACE deletes the note and the instrument
Instrument (TT)	0–F, from \$00 to \$3F. If the note column of the line is empty the column behaves like a note column: you can type a note there
Volume (vV)	0–F. A volume can be written alone, with no note
Speed (FSS)	0–F: a new speed for the song from this line on, from \$01 to \$FF

If the *respect volume* mode (F11) is on, a note typed on a line that has a volume does not replace the volume.

6.2 The length of a track

A track is as long as its *end line*: **CONTROL+END** sets or clears it. **HOME** and **END** go to the start and the end of the track; **CONTROL+HOME** sets the start of a *wise loop*, the line the track goes back to when it reaches its end, so that a repeated phrase is written once. The header of the track shows both numbers (chapter 4).

INSERT and **DELETE** (or **CONTROL+I** and **CONTROL+U**) insert and delete lines in the track.

6.3 Playing while editing

- **ENTER** plays the note at the cursor, **SHIFT+ENTER** plays it and makes its instrument and volume the active ones (a quick way to “pick up” a sound from a song), and **CONTROL+ENTER** plays all the notes of the line.
- **SHIFT+tonekey** plays a note with the active instrument and volume on the current channel, without writing it, and **SHIFT+CONTROL+tonekey** plays it on both stereo channels.
- **F9** switches the current channel off and on, **CONTROL+F9** plays it alone, **CONTROL+1–8** toggle the channels one by one.

6.4 Blocks

A *block* is a selection of lines in a track.

1. Select with SHIFT+UP, SHIFT+DOWN, SHIFT+HOME and SHIFT+END; CONTROL+A selects all the data of the track. ESC deselects.
2. Copy (CONTROL+C), cut (CONTROL+X), paste (CONTROL+V), merge with the content under the cursor (CONTROL+M) and exchange the block with the clipboard (CONTROL+E). With no block the data at the cursor is taken.
3. CONTROL+B restores the block from a backup that is made as soon as the block is started: the way back from a wrong modification.

The blocks can also be changed with keys, either on every line of the block, or only on the lines with the active instrument (SHIFT+CONTROL+A chooses): CONTROL+F1/F2 transpose by semitones, CONTROL+F3/F4 by octaves, SHIFT+CONTROL+LEFT/RIGHT change the instrument and SHIFT+CONTROL+UP/DOWN the volume. The block toolbar has the same buttons.

6.4.1 The Effects and Tools dialog

CONTROL+F (**Block** → **Effects/Tools...**) opens the window of figure 6.1: a choice of effects for all the data of the block, with the fields of the chosen effect. The effects include fading the volume in or out between two levels in percent; **Try** applies the effect, **Restore** takes it back, **Play/Stop** plays the block while you are adjusting it and **Default** sets the fields to their default values.

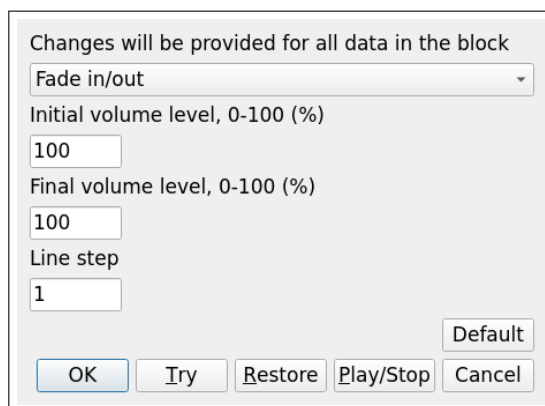


Figure 6.1. The block effects (CONTROL+F).

6.5 Tracks as objects

The **Track** menu works on a whole track: copy, paste, cut, delete, load a track from a file or save it as a text file, ask for information about the current track, *search and build a wise loop* in a track (the program finds the repeats and writes them once) or *expand* it again, and the clean-up operations that remove duplicated or unused tracks from the song. The operations on all the tracks change the whole song; save it before using them.

6.6 Undo

CONTROL+Z undoes the last change and CONTROL+Y does it again. **Edit** → **Clear Undo & Redo History** forgets the changes made so far.

Instruments

An instrument says how a note sounds from the moment it starts: how its volume changes with time, which distortion the POKEY uses, whether the pitch moves. In **INSTRUMENT EDIT** (F3) the instrument with the number in the info area is shown and edited (figure 4.2); **SHIFT+LEFT** and **SHIFT+RIGHT** choose another one, from \$00 to \$3F.

7.1 The parts of an instrument

NAME The name, up to 32 characters.

Envelope

A sequence of steps, one for each frame, from 1 to 32. Each step has the volume (**VOLUME L**, and **VOLUME R** for the right channels of a stereo song), the **DISTORTION** (the POKEY distortion of the step, an even value from 0 to E), a **COMMAND** with its parameters **X** and **Y**, and the flags **AUTOFILTER** and **PORTAMENTO**.

- **ENVELOPE LENGTH**: the number of steps; **ENVELOPE GOTO**: the step to jump to when the end is reached.
- **FADEOUT**: a volume slide that starts when the end of the envelope is reached for the first time (\$00 no slide, \$FF the fastest), down to the volume **VOL MIN**.

Table of notes / frequencies

Up to 32 values added to the note or to the frequency, one for each **TABLE SPEED** frames: arpeggios, pitch bends, drums. **TABLE TYPE** chooses between notes (in semitones) and frequencies; **TABLE MODE** between adding to the base note (**SET**) and adding to the last result (**ADD**). **TABLE LENGTH** and **TABLE GOTO** work as in the envelope.

Effects

DELAY (the number of frames before the effects start), **VIBRATO** (three levels) and **FREQSHIFT** (a frequency shift for each frame).

AUDCTL

The bits of the POKEY control register the instrument sets: the 15 kHz base clock, the two high pass filters, the joined 16-bit channels, the 1.79 MHz clocks and the 9-bit polynomial counter.

The fields, their ranges and the meaning of the envelope commands and of the distortions are in the reference tables of appendix A.

7.2 Editing

The cursor moves with the arrow keys and **TAB**, which jumps between the groups of parameters. The values are typed in hexadecimal (0–F) or changed with **CONTROL+arrow**. In the tables and the envelope, **INSERT** adds a step, **SHIFT+INSERT** duplicates one, **DELETE** removes one and **SPACE** clears one;

HOME and END go to the start and to the end, and CONTROL+HOME and CONTROL+END set the loop and the length.

The volume envelope can be drawn with the mouse: the left button sets the volume of a step, the right one sets it to zero. SHIFT+CONTROL+UP and DOWN move all the steps at once; SHIFT+CONTROL+numblock + and - change the volumes of the left and right envelopes.

Tip

Press SHIFT+ENTER on a note of a track to pick up its instrument and listen to it, and SHIFT plus a note key while the instrument is on the screen to hear the changes as you make them.

7.3 Instrument files and the Instrument menu

The **Instrument** menu has the commands to copy, paste, cut and delete the instrument, to paste only a part of it (only the volume envelopes, only the parameters, only the table...), to show information about the instrument (where it is used), to change all its occurrences in the songs to another number, to renumber all the instruments, to clear the unused ones, to *load* an instrument from a file and to *save* it as an .rti file. The **instruments** folder of the program has files to start from.

7.4 The distortions

The DISTORTION of an envelope step makes the character of the sound. The values are 0 and 8 (white noise), 2 (square-ish tones), 4 (pure by default, and a table of its own in the patched drivers), 6 and C (buzzy bass tones), A (pure tones) and E (gritty bass tones). The exact POKEY registers behind each are in the table of appendix A, and some of the tracker drivers (chapter 15) give some of them a different sound.

7.5 The instrument active help

View → **Instrument Active Help** shows, below the instrument, a description of the value under the cursor in the envelope: what the distortion or the command is, and the effect of its parameters.

The song

8.1 Song lines

The song area (the **SONG EDIT** part, **F4**) lists the song lines. Each column is a channel, and the number in a cell is the track the channel plays on that line. The keys:

- **0–F**, **CONTROL+LEFT**, **CONTROL+RIGHT** and **BACKSPACE** change the track number of the cell.
- **CONTROL+T** puts a new empty track in the cell, **CONTROL+D** makes a copy of the track and puts it in its place (the program asks when the track was used only once), **CONTROL+J** prepares a whole line of empty tracks.
- **INSERT**, **DELETE** (**CONTROL+I**, **CONTROL+U**) insert and delete song lines. The **GO TO LINE** values that point past the change are adjusted by the program.
- **CONTROL+K** inserts a copy or a clone of song lines, in the dialog of figure 8.1.
- **CONTROL+G** makes the line a **GO TO LINE**: the song continues from another song line, which loops it.
- **HOME** and **END**, **SHIFT+PAGE UP** and **SHIFT+PAGE DOWN** move to the start or the end of the song and between the subsongs.

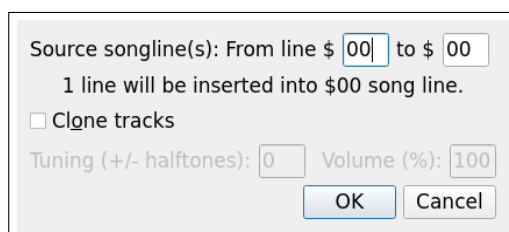


Figure 8.1. Insert a copy or a clone of song lines.

8.2 Subsongs

A song may contain more than one tune: each subsong starts from a song line. The keys **SHIFT+PAGE UP** and **SHIFT+PAGE DOWN** move between them, and the **SAP** export (chapter 12) lists them in its *Subsongs* field: the hexadecimal numbers of the song lines that start the subsongs, the first of which is the default one.

8.3 Bookmarks

F8 sets a bookmark at the cursor and **CONTROL+F8** clears it. **SHIFT+F7** plays from the bookmark, a way to listen to the same passage again and again while you work on it.

8.4 The info area

The INFO EDIT part (SHIFT+F4) edits the data of the song itself.

Field	Meaning
NAME	the name of the song and of the author, up to 64 characters
MUSIC SPEED: AA/MM/S	AA is the current speed of the song, MM the main speed, which the song uses when it is played from the beginning (both from \$01 to \$FF); S is the engine speed, from 1 to 8: how many times the player routine is called in each frame. A speed above 4 is not supported by the exported formats
MAXTRACKLENGTH	the maximal length of the tracks
MONO / STEREO	the number of tracks: 4 or 8
PAL / NTSC	the video system, 50 or 60 frames per second

The values are changed with the hexadecimal keys, or CONTROL+arrow. TAB and LEFT/RIGHT move between the fields, SHIFT+TAB goes to the name and ENTER returns to the part you came from.

8.5 Changing the kind of song

The commands of the **Song** menu that change the whole song open dialogs of their own.

- **Switch song between 4 or 8 channels...** changes a mono song to stereo and the other way round. The data of the channels that disappear is lost, and the dialog says so.
- **Song columns' order change/copy/clear...** (figure 8.2) reorders the columns of the song, copies one into another, turns a mono column into a stereo one or clears a column, for the song lines in a range.
- **Change maximal length of tracks...** (figure 8.3) changes the maximal length of the tracks.
- **All size optimizations...** runs the optimizations that make the song smaller: searching wise loops in all the tracks, clearing the unused ones, ...(section 8.6).

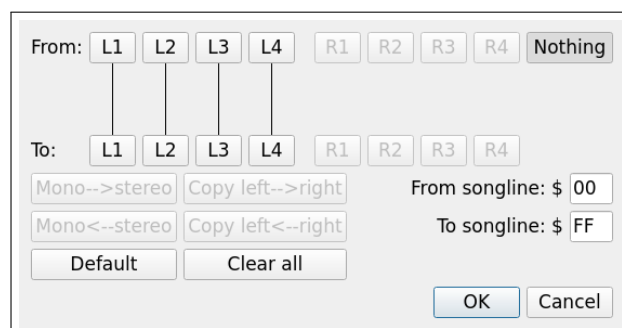


Figure 8.2. The order of the columns of the song.

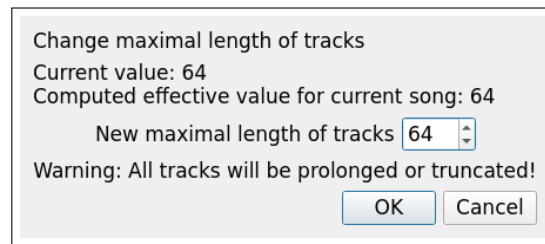


Figure 8.3. The maximal length of the tracks.

8.6 Optimizing the size

The Atari has little memory, so a song should be small. The program has the tools to make it so: the *wise loops* (a track that repeats a phrase is stored once), the clearing of the tracks and the instruments that are not used in the song, the removal of duplicated tracks (the song is adjusted to the one left), and the *cleanup* commands of the **Track** and **Instrument** menus. The export for programmers (*RMT stripped*) shows the features used by the song, which lets a player routine be made smaller still (chapter 12).

Playing and monitoring

9.1 Play commands

Key	Action
F5	play the song from the start
F7	play from the cursor
SHIFT+F7	play from the bookmark
F6	play the current song line in a loop
SHIFT+F6	play the selected block in a loop
ESC	stop and silence
F12	follow mode: the screen follows the playing position
CONTROL+F12	toggle PAL / NTSC
F11	respect volume on/off

The same commands are the buttons of the main toolbar and the entries of the **Play** menu. With the option *Reset the Atari sound routines each time ESC is pressed* (on by default) ESC also starts the player routine again, which clears sounds that hang. A song with the AUTOFILTER that is opened while the Patch16 driver is selected makes the program ask whether to play it with the original RMT 1.28 driver (chapter 15).

9.2 Channels

The **Channels** menu and the keys switch the channels on and off to hear a part alone: F9 mutes or unmutes the active channel, SHIFT+F9 all the channels, CONTROL+F9 plays the active channel alone (solo) and again unmutes everything, CONTROL+1–8 toggle a channel. The mouse does the same on the track headers (chapter 4).

9.3 Watching the sound

- The *play time counter* (**View** → **Play Time Counter**) shows the time and the tempo while the song plays.
- The *volume analyzer* (**View** → **Volume Analyzer**) draws the volume of each channel above the track headers; figure 9.1 shows it.
- **Pokey** → **Pokey Chip Registers** shows the registers of the POKEY with the pitch, volume and distortion of each channel: the real thing the Atari would do.

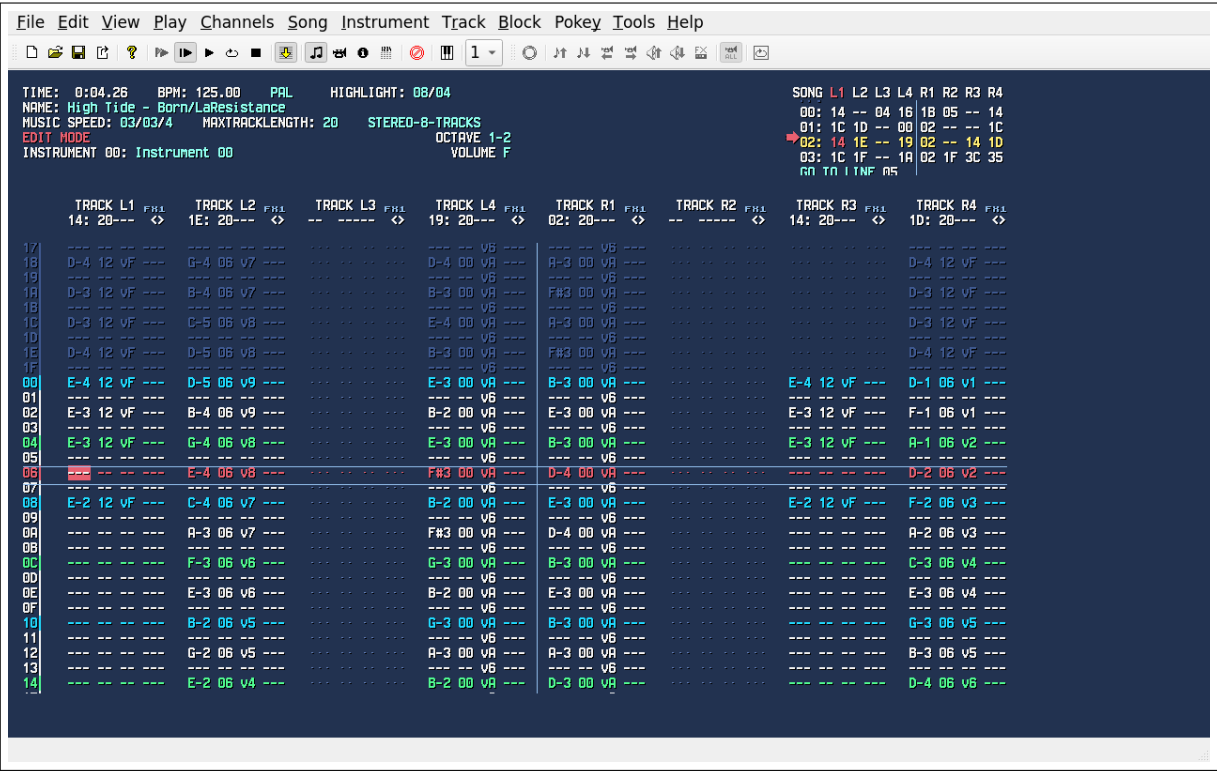


Figure 9.1. The volume analyzer and the registers of the POKEY while a song plays.

9.4 Playing from the keyboard

In prove mode (CONTROL+SPACE) the note keys only play. In edit mode SHIFT plus a note key plays without writing. A MIDI keyboard does the same and can also record (chapter 13).

CHAPTER 10

The menus

This chapter goes through the menu bar. The table of every command with its key and its short description is in appendix A (it is written by the program, so it is always up to date); here each menu is shown and the commands that need an explanation are explained.

10.1 File

New	Ctrl+N
Open...	Ctrl+O
Reopen	Ctrl+R
Save	Ctrl+S
Save As...	Ctrl+Shift+S
Import...	
Export	
Print...	Ctrl+P
Print Preview	
Print Setup...	
Properties	Alt+Return
Exit	Alt+F4

- **New, Open..., Reopen** (loads the song from the last saved version), **Save, Save As...**: the song is an .rmt file; .txt and .rmw are chosen in the file type of the dialog.
- **Import...**: ProTracker modules and TMC songs (chapter 12).
- **Export**: the formats for the Atari (chapter 12).
- **Print, Print Preview, Print Setup**: the screen of the song is printed on a landscape page, scaled to fill it.
- **Properties** (ALT+ENTER): the properties of the song.

10.2 Edit

Undo	Ctrl+Z
Redo	Ctrl+Y
Clear Undo & Redo History	
Edit Tracks	F2
Edit Instruments	F3
Edit Info	Shift+F4
Edit Song	F4
☐ Switch Edit Mode	Ctrl+Space

Undo and redo (the history can be cleared), the four parts of the screen (tracks, instruments, info, song), and the switch between edit and prove mode.

10.3 View

☒ Main <u>T</u> oolbar
☒ <u>B</u> lock Toolbar
☒ <u>S</u> tatus Bar
☒ <u>P</u> lay Time Counter
☐ <u>V</u> olume Analyzer
☒ <u>I</u> nstrument Active Help

Shows or hides the toolbars and the status bar, the play time counter, the volume analyzer and the instrument active help. The choices are kept in the configuration.

10.4 Play

Play From <u>B</u> ookmark	Shift+F7
Play From <u>S</u> tart	F5
<u>P</u> lay	F7
Play And <u>L</u> oop Song Line	F6
<u>S</u> top	Esc
☒ Toggle <u>F</u> ollow Mode	F12

Play from the bookmark, from the start, from the cursor, loop the song line, stop and follow mode (chapter 9).

10.5 Channels

☒ Channel <u>1</u>	Ctrl+1
☒ Channel <u>2</u>	Ctrl+2
☒ Channel <u>3</u>	Ctrl+3
☒ Channel <u>4</u>	Ctrl+4
☒ Channel <u>5</u>	Ctrl+5
☒ Channel <u>6</u>	Ctrl+6
☒ Channel <u>7</u>	Ctrl+7
☒ Channel <u>8</u>	Ctrl+8
<u>M</u> ute/Unmute Active Channel	F9
Mute/Unmute <u>A</u> ll Channels	Shift+F9
<u>S</u> olo Active Channel	Ctrl+F9

Switch on and off each of the channels 1 to 8, mute or unmute the active one or all of them, or play the active one alone.

10.6 Song

C <u>o</u> py Line	
P <u>a</u> ste Line	
C <u>l</u> ear Line	
S <u>e</u> t B <u>o</u> okmark	F8
C <u>l</u> ear B <u>o</u> okmark	Ctrl+F8
D <u>e</u> lete c <u>u</u> rrent line	Ctrl+U
I <u>n</u> sert new empty line	Ctrl+I
I <u>n</u> sert new line with unused empty tracks	Ctrl+J
I <u>n</u> sert c <u>o</u> py or clone of song line(s)...	Ctrl+K
I <u>n</u> sert n <u>e</u> w empty unused track to current song position	Ctrl+T
M <u>a</u> ke a track <u>d</u> uplicate to current song position	Ctrl+D
Switch song to Stereo 8 tracks...	
Song columns' order change/copy/clear...	
Change maximal length of tracks...	
All size optimizations...	

Copy, paste and clear a song line; the bookmark; insert and delete lines; insert tracks; and the commands of chapter 8 that change the whole song.

10.7 Instrument

C <u>o</u> py	
P <u>a</u> ste	
P <u>a</u> ste s <u>pe</u> cial	▶
C <u>u</u> t	
D <u>e</u> lete	
I <u>n</u> fo about current instrument...	
C <u>h</u> ange all the instrument occurences...	
R <u>e</u> number all instruments...	
L <u>o</u> ad instrument from file...	
S <u>a</u> ve instrument as...	
C <u>l</u> ear all unused instruments...	
A <u>l</u> l instruments cleanup...	

Everything to do with an instrument: copy and paste (and paste special), information, change or renumber, clear the unused ones, load and save (chapter 7).

10.8 Track

<u>C</u> opy	
<u>P</u> aste	
<u>C</u> ut	
<u>D</u> elete	
Increase Step Size	Num +
Decrease Step Size	Num -
<u>I</u> nfo about current track...	
Search and <u>b</u> uild wise loop	
<u>E</u> xpand loop	
Search and rebuild wise loops in all tracks...	
Expand loops in all tracks	
Renumber all tracks...	
<u>L</u> oad track from file...	
<u>S</u> ave track as...	
Clear all duplicated tracks, adjust song...	
Clear all tracks unused in song...	
All tracks cleanup...	

Copy, paste, cut and delete a track; the step size; information about the current track; wise loops; renumbering; loading and saving a track; and the clean-up of the tracks of the song.

10.9 Block

Restore From <u>B</u> ackup	Ctrl+B
<u>C</u> opy	Ctrl+C
<u>P</u> aste	Ctrl+V
Paste <u>S</u> pecial	▶
<u>C</u> ut	Ctrl+X
<u>D</u> elete	Del
Exchange Block and Clipboard	Ctrl+E
<u>E</u> ffects/Tools...	Ctrl+F
Select <u>A</u> ll	Ctrl+A

The commands of the blocks (chapter 6): restore from the backup, copy, paste, paste special (merge, only the volumes, only the speeds), cut, delete, exchange, effects and tools, select all.

10.10 Pokey

Pokey Chip Registers

Activate Pokey Explorer Mode

Ctrl+Shift+F5

Channel 1

Channel 2

Channel 3

Channel 4

AUDCTL

SKCTL

Debug Channel

Divisor

The registers of the chip and the Pokey explorer mode, with a sub-menu for each channel (chapter 14).

10.11 Tools

Open ASMA

Open ASAP File...

Run Script...

Options...

Tuning...

Open ASMA opens the Atari SAP Music Archive in the browser; **Run Script...** runs a script (chapter 16); **Options...** and **Tuning...** open the dialogs of chapter 11.

10.12 Help

Help

F1

Online Help

Shift+F1

About RITMO

The online help (a browser page with the documentation) and the **About** window, with the credits.

Options and tuning

11.1 The Options dialog

Tools → **Options...** opens the dialog of figure 11.1. The settings are saved when the program closes, in the settings of the system (chapter 2).

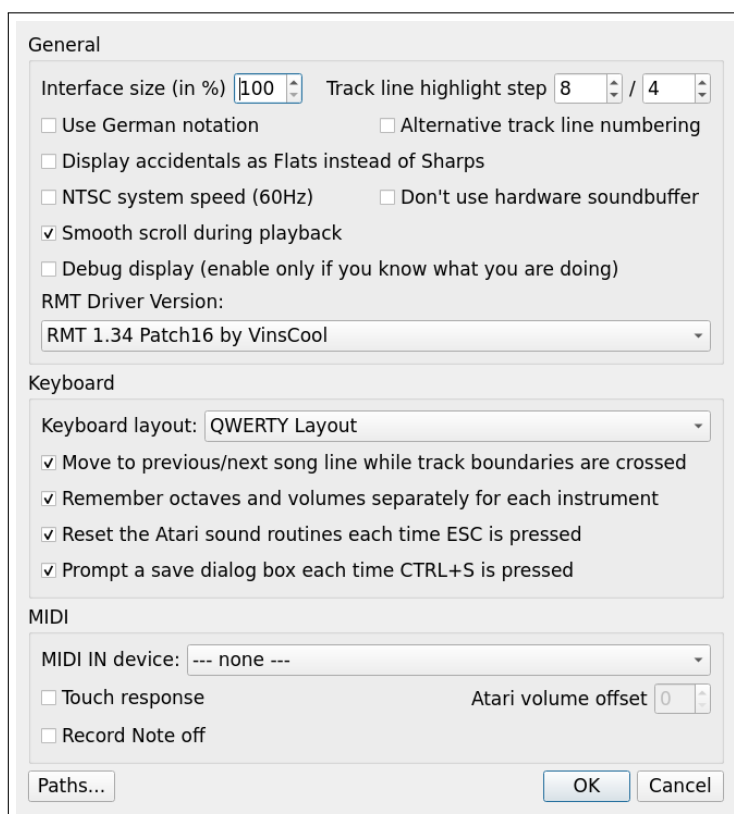


Figure 11.1. The **Options** dialog.

11.1.1 General

Interface size (in %)

The size of the drawing of the window, from 100% to 300%. Only whole multiples of 100% keep the pixels exact.

Track line highlight step

The two numbers of the **HIGHLIGHT** field of the info area: every how many lines the lines of the tracks are highlighted (for example 8 and 4).

Use German notation

The note **B** is written **H**, as in the German notation.

Alternative track line numbering

The line numbers follow the highlight step.

Display accidentals as flats instead of sharps

D# is written as Eb.

NTSC system speed (60 Hz)

The video system of the song: 60 frames a second instead of the 50 of PAL. The same switch is CONTROL+F12.

Don't use hardware soundbuffer

A different way of playing the sound buffer, for the audio systems that need it.

Smooth scroll during playback

The lines of the tracks scroll one pixel at a time while a song plays.

Debug display

A line of internal values in the status bar. Enable it only if you know what it shows.

RMT Driver Version

The tracker driver, the player routine that makes the sound while you work (chapter 15). The driver must be the one the song was made with; the default is *RMT 1.34 Patch16 by VinsCool*.

11.1.2 Keyboard

Keyboard layout

QWERTY, QWERTZ or AZERTY: where the piano of the note keys is. The first start picks the layout from the language of the keyboard.

Move to previous/next song line while track boundaries are crossed

The cursor passes from the end of a track to the next song line, and the other way round.

Remember octaves and volumes separately for each instrument

Each instrument keeps the octave and the volume you last used with it.

Reset the Atari sound routines each time ESC is pressed

ESC also starts the player routine again.

Prompt a save dialog box each time CTRL+S is pressed

CONTROL+S asks where to save, also from the menu and the toolbar.

11.1.3 MIDI

The MIDI input device, the touch response, the Atari volume offset and *Record note off*: they are described in chapter 13.

11.1.4 Paths

The **Paths...** button opens a dialog with the default folders of the songs, the instruments and the tracks, each with a **Browse** button. The dialogs to open and save start there.

11.2 The tuning

Tools → **Tuning...** (figure 11.2) is a calculator for the pitch of the notes. The POKEY makes its frequencies by dividing a clock, so not every pitch exists: the program computes the divisors that come nearest to the notes of the temperament you choose, and the tracker plays with them.

General		
Base tuning (Hz)	440.83751645933	Base note A-
eg: 440, 440.83751645933, 432, 443.9, 444.895778867913, etc		
Temperament	Equal Temperament (Default)	
RA/TIO		
Unison	1	/ 1
Minor 2nd	20	/ 19
Major 2nd	10	/ 9
Minor 3rd	20	/ 17
Major 3rd	5	/ 4
Perfect 4th	4	/ 3
Tritone	60	/ 43
Perfect 5th	3	/ 2
Minor 6th	30	/ 19
Major 6th	5	/ 3
Minor 7th	30	/ 17
Major 7th	15	/ 8
Octave	2	/ 1

Figure 11.2. The **Tuning** dialog.

Base tuning (Hz) and Base note

The frequency of a reference note, by default A- at 440.8375... Hz. It can be any number: 432, 443.9...

Temperament

Equal Temperament (the default) or another. The *Ratio* table lists the interval of each step over the unison as a fraction.

Reset

puts the default values back; **Test now** tries the tuning.

The tuning is saved with the configuration and is also the one a script uses when it exports a song (chapter 16).

Import and export

12.1 Importing

File → **Import...** reads a ProTracker module (.mod) or a TMC song (.tmc, .tm8) and makes a song of it. After the file is chosen the dialog of figure 12.1 asks how to convert a module.

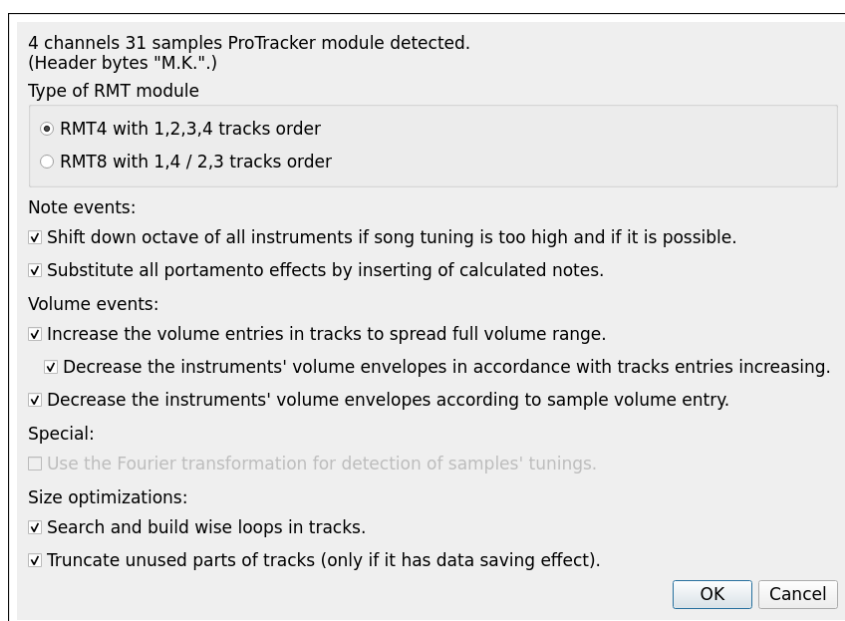


Figure 12.1. Importing a ProTracker module.

- The *type of RMT module*: **RMT4** (the four channels of the module on the four tracks) or **RMT8** (the channels spread on the two POKEYs, in the order 1, 4 / 2, 3).
- The treatment of the *notes* (shift the octave down if the tuning of the song is too high, replace the portamento effects by calculated notes), of the *volumes*, and the *size optimizations* that are made while importing (wise loops, cutting the unused parts of the tracks).

When the conversion is done a dialog reports how many tracks, instruments and song lines the song has, and what the optimizations did. Check the box and press **OK**: the song is loaded, but it still needs your work. The instruments are rough: look at their envelopes and distortions (drums, basses...) and correct the tunings of the original samples.

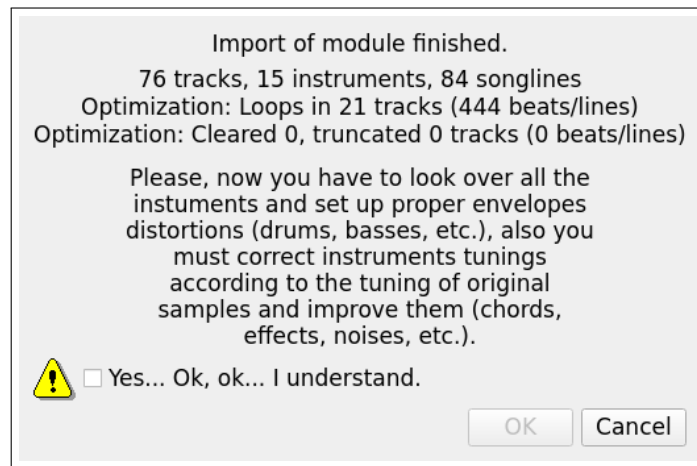


Figure 12.2. The result of the import.

12.2 Exporting

File → **Export** writes the song in a format for the Atari. The file dialog chooses the name and the format; most of the formats then show a dialog of options. Every format is written from the data of the song as the tracker plays it.

Format	Use
RMT stripped song file (.rmt)	the module without the names and the unused tracks and instruments, to link in a program that uses the RMT player routine
ASM simple notation (.asm)	the notes as assembler data, for your own player
SAP-R data stream (.sapr)	the register writes of the POKEY, frame by frame
Compressed SAP-R (.lzss)	the same, compressed, for the LZSS player
SAP file + LZSS driver (.sap)	a file the Atari players (ASAP, the plug-ins of many players, ASMA) play directly
XEX Atari executable (.xex)	a program for the Atari that plays the song when it is started
Relocatable ASM for RMTPlayer (.asm)	the module as source with labels, to be placed in memory by your program
WAV audio file (.wav)	the sound as a 16-bit, 44.1 kHz, stereo file, the song played once up to its loop

12.2.1 RMT stripped

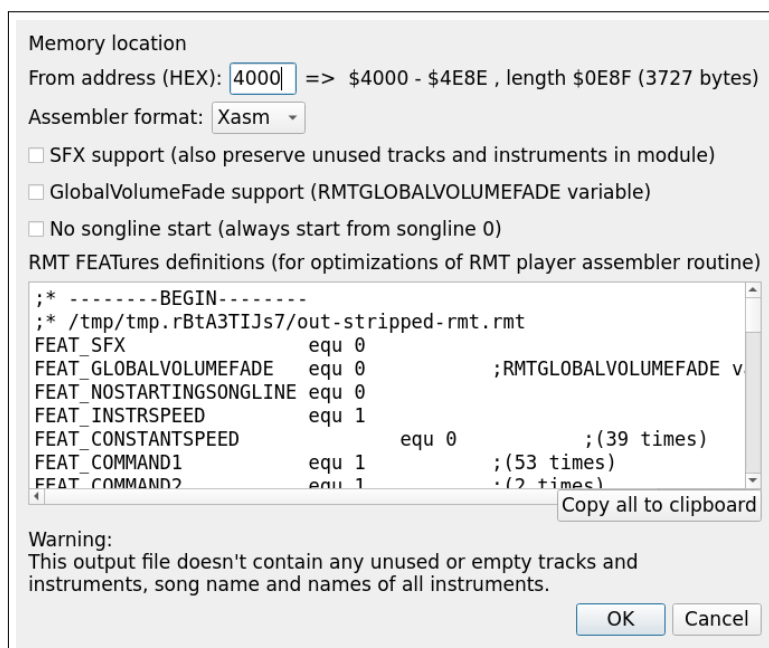


Figure 12.3. The export of an RMT stripped file.

The *from address* is where the module will be in the memory of the Atari; the program shows the range it takes. The options are *SFX support* (keep the unused tracks and instruments, for sound effects played by the game), *GlobalVolumeFade* (a variable the program can use to fade the music) and *No songline start* (the song always starts from song line 0). The text box lists the *RMT FEATures* the song uses: if a game uses only this song, these definitions can be pasted into the player routine source (`rmt_feat.a65`), which then assembles to a smaller and faster routine (this is described in the file `optimization.txt` that comes with the program).

12.2.2 ASM simple notation

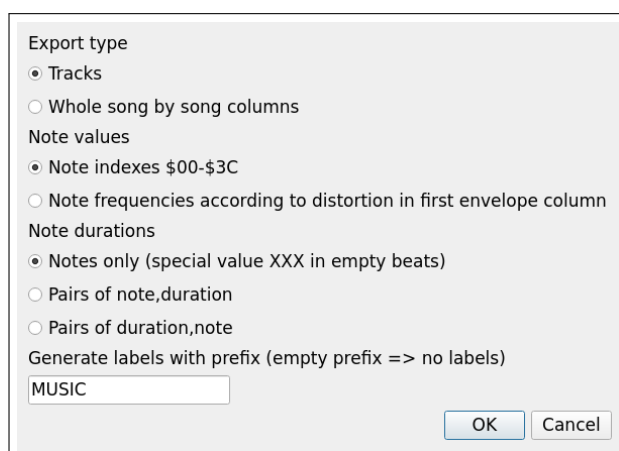


Figure 12.4. The export of the notes as assembler data.

Choose what to export (the tracks, or the whole song by song columns), how the note is written (the index \$00–\$3C, or the frequency according to the distortion of the instrument), how the duration is written (notes only, with the special value XXX in empty beats, or pairs of note and duration, in either order) and the prefix of the labels (empty for none).

12.2.3 SAP, SAP-R and LZSS

Figure 12.5. The export of a SAP file.

A SAP file has a header with the name, the author and the date, which the dialog fills with those of the song. The field *Subsongs* lists, in hexadecimal, the song line each subsong starts from: 00 for one tune, 00 10 for two. SAP-R is the stream of the POKEY registers; LZSS compresses it, and the SAP file contains it with the player routine of the VU-Player.

12.2.4 XEX

Figure 12.6. The export of an Atari executable.

The program an XEX file contains shows a text on the Atari screen while it plays: up to five lines of 40 characters (the fifth shown instead of the fourth when the SHIFT key of the Atari is held down). Below, the *rasterbar* shows the CPU usage of the player in colour, *shuffle* changes its colours, and the box *Automatically adjust playback speed*, with the line that says that the speed

will be adjusted to 50 Hz on both PAL and NTSC systems, turns the adjustment of the speed on. The preview shows the text as the Atari will draw it.

12.2.5 Relocatable ASM

The RMT song will be exported as byte definitions with specific labels.
The code is fully relocatable and can be split over various locations in memory!

ASM Label for start of song data:

☐ Relocate instruments to ASM Instruments Label:

☐ Relocate tracks to ASM Tracks Label:

☐ Relocatable song lines to ASM Song Label:

Assembler format:

☐ SFX support (also preserve unused tracks and instruments in module)

☐ GlobalVolumeFade support (RMTGLOBALVOLUMEFADE variable)

☐ No songline start (always start from songline 0)

Length \$0E8F (3727 bytes)

Header	= \$00b8 (184) bytes
Instruments	= \$01b8 (440) bytes
Tracks	= \$0bbf (3007) bytes
Song Lines	= \$005e (94) bytes

Sequence of modules:

- Header
- Instruments
- Tracks

OK Cancel

Figure 12.7. The export of a relocatable module.

The module is written as byte definitions with labels (the one of the start of the song data, and optionally separate labels for the instruments, the tracks and the song lines, which can be placed in different places of memory). The text box shows the size of each part and the order of the modules.

12.2.6 WAV

The WAV export has no options: it plays the song with the built-in emulation and writes the sound.

12.3 Exporting from the command line

All the formats can be exported from a script, with no window, so that the files of a release are made from the .rmt source (chapter 16).

MIDI input

RITMO plays and records notes from a MIDI keyboard. The input device is chosen in **Tools** → **Options** (the MIDI group); the button **Toggle MIDI on/off** of the toolbar switches the input on and off while a device is set. The device is a port of the system: on Linux the ports of the ALSA sequencer (a USB keyboard or a virtual port), on Windows WinMM, on macOS CoreMIDI.

13.1 Options

Option	Meaning
MIDI IN device	the input device, or <i>none</i>
Touch response	the key velocity becomes the volume of the note: Atari Volume Offset + velocity / 8 , limited to 1–15. Without it the current volume (as for the typed notes) is used
Atari volume offset	added to the volume of the velocity, from 0 to 15
Record note off	a released key (a note off, or a note on with velocity 0) deletes the note it played and writes volume 0. Without it releases are ignored

13.2 The MIDI channels

Channel	Function
1	Records notes at the cursor, like the note keys: the MIDI note 36 is the lowest note of the tracker (C-1), the note is written with the active instrument, the cursor moves down by the note spacing, and the note is played. While the song plays with follow mode, a note in the first half of a line is written on the next line (quantized). Outside the tracks area, in a jam mode or with SHIFT or CONTROL held down, the note is only played. Recording needs the window to have the focus
2–9	Live play on the channels L1–L4 and R1–R4 (6–9 repeat L1–L4 on a mono song): never recorded, and independent of the focus. The velocity is the volume (velocity / 8); a program change sets the instrument of the channel; the controller 123 (all notes off) stops its note and the controller 121 (reset all controllers) restarts the player routine
11–15	a program change chooses the active instrument
10 and 16	an experimental assignment of controllers, kept from the original program, with the POKEY explorer (chapter 14) on channel 16

A system reset message (**FF**) restarts the player routine and forgets the notes held on the channels 2 to 16. The implementation chart of the MIDI messages is in appendix C.

13.3 Scripts

A script can send MIDI messages to the program with the `midi` command, which is how the recording can be tested without a keyboard (chapter 16).

The Pokey explorer

The explorer is a mode to study the sound of the POKEY. **Pokey** → **Activate Pokey Explorer Mode** (CONTROL+SHIFT+F5) stops the tracker's player routine and lets the keys write the registers of the chip directly, so that you can hear the effect of each divisor, distortion and AUDCTL bit. The mode is left with the edit/jam toggle (CONTROL+SPACE).

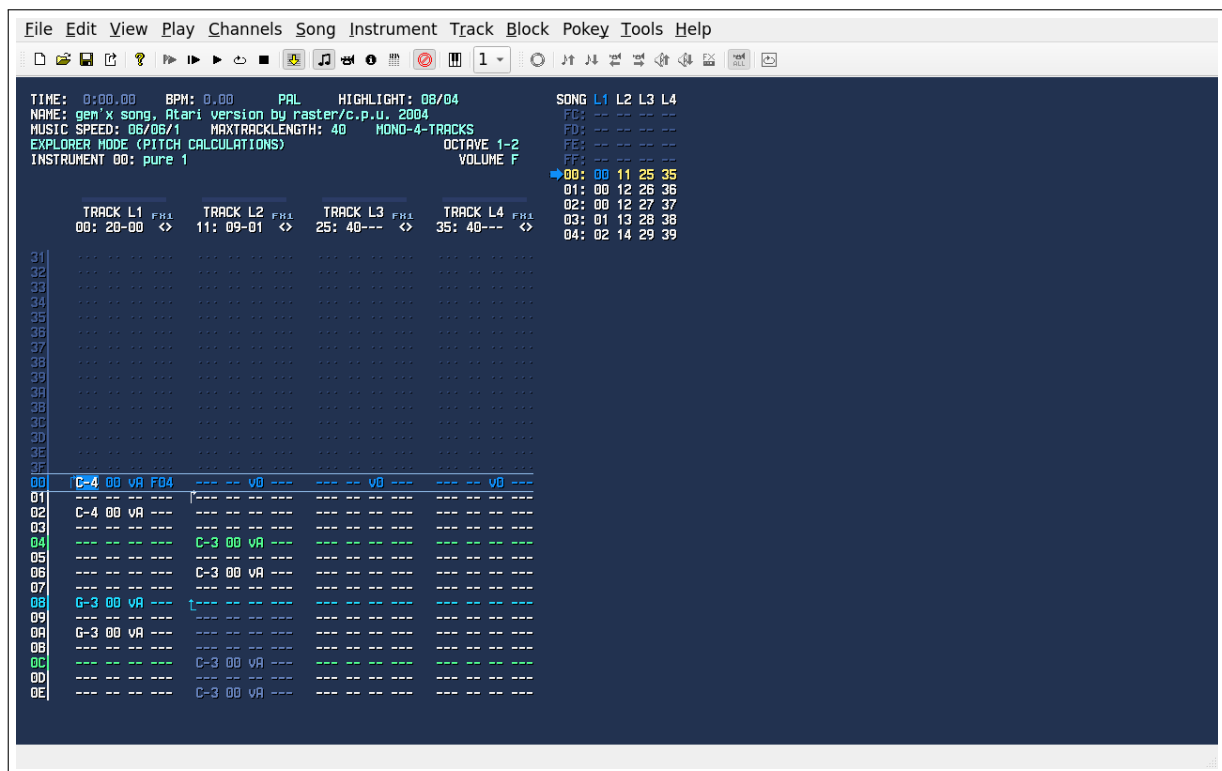


Figure 14.1. The Pokey explorer mode, with the registers of the chip on the screen.

With **View** → **Pokey Chip Registers** on, three rows under the dumps of the registers detail one channel: its AUDF and AUDC bytes, the first divisor of AUDF + MODOFFSET from 3 up (MODULO), the coarse divisor of the pitch formula (28 at 64kHz, 114 at 15kHz, 1 at 1.79MHz and for joined channels), the free divisor, the modulo offset (1, 4 at 1.79MHz, 7 for joined channels) and the resulting pitch.

14.1 The keys

The keys are the physical positions of the QWERTY layout (on QWERTZ and AZERTY they are the keys at the same positions).

Key	Function
ENTER / BACKSPACE	next / previous channel (0–3)
+ / -	divisor +0.1 / -0.1; with SHIFT +1.0 / -1.0 (from 1.0 to 10000.0)
1 3 5 7	AUDF0 to AUDF3 +1; with SHIFT +\$10
Q E T U	AUDF0 to AUDF3 -1; with SHIFT -\$10
2 4 6 8	AUDC0 to AUDC3 +1; with SHIFT +\$10
W R Y I	AUDC0 to AUDC3 -1; with SHIFT -\$10
C G F K J D A P	toggle the bits 0 to 7 of AUDCTL: the 15 kHz clock, the two high pass filters, the two 16-bit joins, the two 1.79 MHz clocks and the 9-bit poly counter
M	toggle the two-tone bits of SKCTL

The **Pokey** menu has the same operations as menu items, which work in the explorer mode only.

Tracker drivers

15.1 Tracker Drivers

A tracker driver is part of the tracker that generates the actual sound from the tracker data and user input. In the case of RMT, the tracker driver is a piece of MOS 6502 code that runs on an emulated Atari 8-bit computer.

15.2 Tracker Driver Versions

A tracker driver version is a variant of the tracker driver that interprets the data and user input differently. For example, different tracker driver versions have a distinct feature set or tuning. A correct replay of the music requires using the same tracker driver version that was used to create it.

The original tracker driver version is the one by Raster in RMT 1.28. Many people modified the driver code and data to yield new effects. All known older versions are preserved for compatibility.

Tracker driver versions are identified in different ways. For inclusion into RMT, the tracker driver versions are numbered starting from 1 (0 is no driver). The number is the one of `TRACKERDRIVERVERSION` in `rmt.ini`, of the constant in the source code (`TrackerDriverVersion.h`) and of the binary `resources/drivers/rmt_driver_v<code>.obx`. Additionally, they have a readable name for users (the "RMT Driver Version" list of the Options dialog) and a name for scripts (`set driver`, see [rmt_scripting.md](#)).

The level of a driver is the RMT version whose driver code it is: the original by Raster or a patch of it.

Code	Name in the Options dialog	Level (driver code)	Script name	Constant name	Author
0	No RMT Driver	-	-	NONE	-
1	RMT 1.28 Unpatched by Raster	RMT 1.28, original	unpatched	UNPATCHED	Raster
2	RMT 1.28 Unpatched with Tuning	RMT 1.28, original (same binary as 1)	unpatched-with-tuning	UNPATCHED_WITH_TUNING	VinsCool
3	RMT 1.25 Patch3 Instrumentarium by Analmux	RMT 1.25, Patch 3	patch3	PATCH3	Analmux
4	RMT 1.27 Patch6 by Analmux	RMT 1.27, Patch 6	patch6	PATCH6	Analmux
5	RMT 1.28 Patch8 by Analmux	RMT 1.28, Patch 8	patch8	PATCH8	Analmux
6	RMT 1.34 Patch16 by VinsCool	RMT 1.28, Patch16-3.2 (RMT 1.31 and later; the default since RMT 1.34)	patch16	PATCH16	VinsCool

Code	Name in the Options dialog	Level (driver code)	Script name	Constant name	Author
7	RMT 1.28 Patch Prince of Persia by VinsCool	RMT 1.28, Patch 8 modified	patch-prince-of-persia	PATCH__PRINCE__OF__PERSIA	VinsCool

The default is 6, Patch16.

Compatibility with the AUTOFILTER of RMT 1.28 (measured on a song that uses the AUTOFILTER: the frequency offset of channel 3 against channel 1 is the same as with the original RMT 1.28 player):

Code	AUTOFILTER as RMT 1.28
1, 2, 3, 4, 5, 7	yes
6	no (see RMT 1.34 Patch16 by VinsCool)

15.2.1 RMT 1.28 Unpatched by Raster

The original RMT driver version that set the standard for many years with Atari 8-bit music. This version was mainly with RMT 1.28, as well as the unofficial RMT 1.30 by Rudla.

15.2.2 RMT 1.28 Unpatched with Tuning

This driver version by VinsCool was never completed: the binary `rmt_driver_v2.obx` is the same as `rmt_driver_v1.obx`, so it plays exactly as RMT 1.28 Unpatched. It is listed so that a configuration or a script that names it keeps working.

15.2.3 RMT 1.25 Patch3 Instrumentarium by Analmux

Adapted from Analmux's original post on [AtariAge](#): > The music track "Instrumentarium" is composed, tracked, and RMT-patched by me: ANALogue MUltipleXer. I used a patch to change some RMT possibilities: f.e, I added sawtooth wave (ringmod), triangle wave (linear subharmonic), clarinet (16-bit), distortion guitar (16-bit), and sync modes (2-tone filter & 16-bit). At the end of 2009, this tune was named "Instrumentarium Remix 1"; it's the same tune. I was using exotic undocumented POKEY features. It takes almost no CPU time.

15.2.4 RMT 1.27 Patch6 by Analmux

Adapted from Analmux's original post on [AtariAge](#): > Tables `freqtabbasslo` and `freqtabbasshi` are patched (all I did was multiply each 16-bit value by 15). The '16-bit distortion 6' instrument, which was really a generator C instrument, is now a generator A instrument. I removed the blocking of the 1st channel, to enable 16-bit filter-like effects."

15.2.5 RMT 1.28 Patch8 by Analmux

Adapted from Analmux's original post on [AtariAge](#):

RMT Distortion	POKEY Distortion	Bits	Clock	Number of Bytes	Instrument
0	0	8	x	x	White noise
2	2	8	1.79 MHz	48	Poly 5 / Generator 2
4	A	8	1.79 MHz	36	Sawtooth / Triangle
6	C	8LSB	1.79 MHz	32	Clarinet (Poly 4), LSB
8	A	8HSB	1.79 MHz	64	Harmonic square undertones of Clarinet / Distortion guitar, HSB
A	A	8	15 kHz	48	Pure bass
A	A	8	1.79 MHz	12	Pure flute
C	C	8	1.79 MHz	31	Poly 4 (degenerate 1)
C	C	8	1.79 MHz	17	Poly 4 (degenerate 3)
E	8	8LSB	1.79 MHz	32	Distortion guitar (Poly 9), LSB

15.2.6 RMT 1.34 Patch16 by VinsCool

This driver version is an extensive hack of the original RMT 1.28 driver by Raster, inspired directly by Analmux's RMT Patch8 hack. The new features include: more tuning tables, a two-tone filter toggle, an AUDCTL timing fix, AUTOFILTER behavior tweaks, etc. Unfortunately, this patch also introduced many compatibility issues with older modules and a couple of bugs. Despite that, this is probably the most "feature-rich" version of the RMT driver patches created so far. All that came at the cost of extra memory usage and heavier CPU load, so this is no longer an "optimal" driver. This is currently the default RMT driver version included and used by RMT 1.34 and later. The driver code is version Patch16-3.2 (`asm/Patch-16/rmtplayr.a65`), used by RMT since version 1.31; the name in the Options dialog says RMT 1.34, the version that made it the default.

The AUTOFILTER is one of these compatibility issues: Patch16 starts the frequency offset of channel 3 at 0 instead of 1 and does not accumulate the command 6, so the two channels of the filter stay in phase and an older song sounds quiet and an octave higher. A song file does not say which driver it was made with, so when the driver is Patch16 and a song that uses the AUTOFILTER is opened, RMT asks whether to play it with the RMT 1.28 Unpatched driver. The answer is for that song only: a new song or the next song opened uses the driver of the Options dialog again. A script (`rmt /SCRIPT:`) answers No; it chooses the driver with `set driver`.

15.2.7 RMT 1.28 Patch Prince of Persia by VinsCool

This driver version is an older patch based on the original RMT 1.28 Patch 8 by Analmux. This version was created exclusively for the Prince of Persia soundtrack. It was thus never really used outside of a very few test tunes that used some of the sounds.

Scripting

RASTER Music Tracker can run a script file from the command line: open a song, set what the export dialogs would ask, export in any format, save. This is meant for batch files, build scripts and CI, where the SAP, XEX, WAV and ASM files of a song are produced from its .rmt source reproducibly, and for repeated export sets while composing. The script format is the one of the Windows `Ritmo.exe` and of the Java port of the RASTER-Music-Tracker repository, so the same script runs in all of them.

16.1 Running a script

```
ritmo /SCRIPT:build.rmtscript
```

`-script:<file>` and `--script=<file>` are accepted too. The program starts without showing its window and exits with the script's exit code. The configuration (`rmt.ini` and `tuning.ini`, kept in the user's settings - `~/.config/ritmo-atari.org/ritmo.conf` on Linux) is read as for the window, so the tuning shapes the exports exactly as in the tracker; it is not written back.

- **Linux:** no display is needed - without `QT_QPA_PLATFORM` set, a script run uses Qt's offscreen platform. Nothing is played: the sound output is not opened.
- **Windows:** `rmt.exe` is a windowed program, so `cmd.exe` returns to the prompt at once unless it is started with `start /wait` (in PowerShell: `Start-Process rmt.exe "/SCRIPT:build.rmtscript" -Wait`).
- **macOS:** `Ritmo.app/Contents/MacOS/ritmo /SCRIPT:build.rmtscript`.

Messages that would be boxes in the window are printed to the console: errors and warnings to `stderr`, information to `stdout`; a question is answered "No". On Windows, when there is no console (started from Explorer or a scheduler), they go to `<script>.log` next to the script.

Exit codes of the command line:

Code	Meaning
0	every command succeeded
1	a command failed; the script stopped there and the message names the line
2	the script file could not be read or has a syntax error (the message names the line)

16.2 Syntax

- One command per line. Blank lines are ignored, `#` starts a comment (at the start of a line or before a token).
- Tokens are separated by blanks. A value with blanks is quoted: `"My Song.sap"`. Inside

quotes, \" is a quote and \\ a backslash.

- Options are written `name=value` after the command's arguments, in any order: `export sap out.sap author="Raster" subsongs=0`. A quoted value works after the = as well.
- Command names, option names and option values that are choices (`yes/no`, format names, driver names) are case-insensitive; other values (texts, labels, file names) keep their case.
- Yes/no options accept `yes`, `no`, `true`, `false`, `on`, `off`, `1`, `0`.
- Relative file names are relative to the script file's folder, so a script travels with its songs. Output files (`save`, `export`) go to the folder of `set output` instead when the script gave one; the folder is created when it is missing.
- An output file that exists already is an error, unless the script says `set overwrite yes` first.

16.3 Commands

Command	Meaning
<code>open <file></code>	Opens a song. The format follows the extension: <code>.rmt</code> , <code>.txt</code> or <code>.rmw</code> (as File > Open).
<code>save <file></code>	Saves the song as <code>.rmt</code> , <code>.txt</code> or <code>.rmw</code> by extension (as File > Save as). The RMT format's integrity check applies.
<code>export <format> <file> [name=value ...]</code>	Exports the song (as File > Export as). Formats and options below.
<code>set overwrite yes no</code>	Whether later <code>save/export</code> commands may replace existing files. Default <code>no</code> .
<code>set output <folder></code>	Where later <code>save/export</code> commands write (relative to the script's folder; created when missing). Default: the script's folder.
<code>set ntsc yes no</code>	The Options dialog's NTSC setting (the video standard: 60 or 50 frames per second, and the POKEY clock).
<code>set driver <version></code>	The Options dialog's tracker driver: <code>unpatched</code> , <code>unpatched-with-tuning</code> , <code>patch3</code> , <code>patch6</code> , <code>patch8</code> , <code>patch16</code> (the default), <code>patch-prince-of-persia</code> .
<code>set midi-touch-response yes no</code>	The Options dialog's MIDI "Touch response" (the velocity becomes the volume). Default: <code>rmt.ini</code> 's.
<code>set midi-volume-offset <0-15></code>	The Options dialog's MIDI "Atari Volume Offset". Default: <code>rmt.ini</code> 's.
<code>set midi-note-off yes no</code>	The Options dialog's MIDI "Record note off". Default: <code>rmt.ini</code> 's.
<code>midi <status> <data1> [<data2>]</code>	Handles a MIDI message given as hex bytes exactly as one from the MIDI IN device would be (see the manual's "MIDI Input"): <code>midi 90 3C 64</code> is note on C-3, velocity 100, on channel 1 and records the note at the cursor; <code>midi CA 05</code> selects instrument 5. The window counts as focused.

Command	Meaning
<code>dump actions <file></code>	Writes the program's command table - every menu item and toolbar button with its key and its prompt from <code>Rmt.rc</code> - as a Markdown table. The table is read from the running program's menus and toolbars, so it lists the keys the menu items really have (the tracker's own keys, handled in the editing code, appear only where a menu label shows them). A row marked <code>ERROR</code> is a menu label that shows another key than the item has and fails the command.
<code>dump notekeys <file></code>	Writes the note keys of the QWERTY, AZERTY and QWERTZ keyboard layouts (which key plays which note) as keyboard pictures and Markdown tables .
<code>echo <text ...></code>	Prints the text.
<code>quit</code>	Ends the script (implicit at its end).

16.4 Export formats and options

The format names are those of the Export dialog's file-type list. Every option has the value the dialog would show when it opens, so a bare `export sap out.sap` produces exactly what the dialog's OK produces with untouched fields. The extension of the format is added to the file name when it is missing.

Format	File	Options
<code>stripped-rmt</code>	<code>.rmt</code>	<code>address</code> (the module's address, <code>\$4000</code> or decimal; default: the loaded module's), <code>sfx</code> (SFX support), <code>gvf</code> (global volume fade), <code>nos</code> (no starting songline), <code>asmformat = xasm</code> or <code>atasm</code>
<code>asm</code>	<code>.asm</code>	<code>type = tracks</code> or <code>song</code> (tracks, or the whole song by song columns), <code>notes = index</code> or <code>freq</code> , <code>durations = notes</code> , <code>note-duration</code> or <code>duration-note</code> , <code>prefix</code> (the label prefix, empty = no labels; default <code>MUSIC</code>)
<code>sapr</code>	<code>.sapr</code>	<code>author</code> , <code>name</code> , <code>date</code> (default: the song's name, the song's author, today), <code>subsongs</code> (default: the subsongs found in the song)
<code>lzss</code>	<code>.lzss</code>	none. Writes the full tune and, when they are longer than 16 bytes, the <code>_INTRO.lzss</code> and <code>_LOOP.lzss</code> siblings.
<code>sap</code>	<code>.sap</code>	as <code>sapr</code> (a SAP of type B with the LZSS player of VU-Player V2; each songline listed in <code>subsongs</code> starts a subsong, <code>subsongs="00 10"</code> makes two)
<code>xex</code>	<code>.xex</code>	<code>text</code> (the screen text, up to 5 lines of 40 characters, <code>\n</code> between lines; default: the song's name, date and author lines the dialog proposes), <code>rasterbar</code> (yes/no), <code>shuffle</code> (yes/no), <code>region-auto</code> (yes/no), <code>color</code> (the rasterbar color, 0-255)

Format	File	Options
rmtplayer-asm	.asm	startlabel (default RMT_SONG_DATA), relocate (a comma list of instruments, tracks, songlines, or none), instruments-label, tracks-label, songlines-label, asmformat, sfx, gvf, nos
wav	.wav	none (16-bit, 44.1 kHz, 2 channels, the song once up to its loop point)

Options that a format does not have, and values that are not valid, stop the script with a message naming the line.

16.5 Example

```
# Build the release assets of Delta
set overwrite yes
open Delta.rmt
export sap Delta.sap author="Raster" name="Delta" subsongs="0"
export xex Delta.xex text="DELTA BY RASTER\nJAVA PORT" rasterbar=no
export stripped-rmt Delta_stripped.rmt address=$5000
export wav Delta.wav
echo finished
```

```
ritmo /SCRIPT:Delta/build.rmtscript
```

16.6 Environment variables

Two variables let a build script run one and the same script file with different outputs, without editing it:

Variable	Programs	Meaning
RMT_SCRIPT_OUTPUT=<folder>	all	The output folder for every save/export; overrides the script's set output.
RMT_SCRIPT_LOG=<file>	all	Writes the messages to this file instead of the console or <script>.log.
RMT_SCRIPT_SHOW_WINDOW=1	all	Shows the window during the script run (normally it stays hidden), to watch what a script does.

16.7 Tests

test-resources/scripts holds two reference scripts (delta.rmtscript: the mono reference song through every save and export, once with the defaults and once with every option set; midi.rmtscript: MIDI messages recorded into it). scripts/test-scripting.sh [path/to/ritmo] runs both twice into separate folders and checks the exit codes, that the files exist and that the

two runs are byte-identical; it also checks the exit codes 1 and 2. The scripts are the ones the Windows program and the Java port compare their exports with, so the output folders can be compared with those programs' as well (WAV files excepted: 8-bit in `Ritmo.exe`, 16-bit in the Java port).

16.8 Not available as script commands

Importing (MOD, TMC), the instrument files (`.rti`), the track files and `New` - the tracker's interactive functions - and, in this program, "Tools > Run Script" in the window. The command set follows the export workflow; further commands can be added if a need arises.

APPENDIX A

Reference of the commands, keys and fields

This appendix is the complete reference: every command of the menus and the toolbars with its key, the note keys of the three keyboard layouts, and the keys and the fields of each part of the screen. The first two tables are written by the program itself.

Every command of the menus and toolbars, with its key and what it does. This table is generated from the program itself (the `dump actions` script command, see the scripting documentation; `scripts/make-docs.sh` regenerates it), so it is always the current state of the program.

Access Path	Entry	Accelerator Key	Action
Menu File / Tool Bar Main	New	Ctrl+N	Create a new song
Menu File / Tool Bar Main	Open...	Ctrl+O	Load an existing song
Menu File	Reopen	Ctrl+R	Reload song from the last saved version
Menu File / Tool Bar Main	Save	Ctrl+S	Save the song
Menu File	Save As...	Ctrl+Shift+S	Save the song with a new filename
Menu File	Import...		Import song from file
Menu File / Tool Bar Main	Export		Export song to file
Menu File	Print...	Ctrl+P	Print the active view
Menu File	Print Preview		Preview the page
Menu File	Print Setup...		Change the printer and printing options
Menu File	Properties	Alt+Enter	Display the properties of the song
Menu File	Exit	Alt+F4	
Menu Edit	Undo	Ctrl+Z	Undo the last action
Menu Edit	Redo	Ctrl+Y	Redo the last action
Menu Edit	Clear Undo & Redo History		
Menu Edit / Tool Bar Main	Edit Tracks	F2	Move to track edit view

Access Path	Entry	Accelerator Key	Action
Menu Edit / Tool Bar Main	Edit Instruments	F3	Move to instrument edit view
Menu Edit / Tool Bar Main	Edit Info	Shift+F4	Move cursor to Info edit
Menu Edit / Tool Bar Main	Edit Song	F4	Move cursor to song edit
Menu Edit / Tool Bar Main	Switch Edit Mode	Ctrl+Space	Edit/Jam mode toggle
Menu View	Main Toolbar		Show or hide the toolbar
Menu View	Block Toolbar		
Menu View	Status Bar		Show or hide the status bar
Menu View	Play Time Counter		
Menu View	Volume Analyzer		
Menu View	Instrument Active Help		
Menu Play / Tool Bar Main	Play From Bookmark	Shift+F7	Play song from bookmark position
Menu Play / Tool Bar Main	Play From Start	F5	Play song from start position
Menu Play / Tool Bar Main	Play	F7	Play song from current position
Menu Play / Tool Bar Main	Play And Loop Song Line	F6	Play and loop current tracks pattern
Menu Play / Tool Bar Main	Stop	Esc	Stop playing the song. Mute all sounds.
Menu Play / Tool Bar Main	Toggle Follow Mode	F12	Follow the currently playing position (turn on/off)
Menu Channels	Channel 1	Ctrl+1	
Menu Channels	Channel 2	Ctrl+2	
Menu Channels	Channel 3	Ctrl+3	
Menu Channels	Channel 4	Ctrl+4	
Menu Channels	Channel 5	Ctrl+5	

Access Path	Entry	Accelerator Key	Action
Menu Channels	Channel 6	Ctrl+6	
Menu Channels	Channel 7	Ctrl+7	
Menu Channels	Channel 8	Ctrl+8	
Menu Channels	Mute/Unmute Active Channel	F9	
Menu Channels	Mute/Unmute All Channels	Shift+F9	Mute / Unmute all channels
Menu Channels	Solo Active Channel	Ctrl+F9	Mute all channels except active channel / Unmute all channels
Menu Song	Copy Line		
Menu Song	Paste Line		
Menu Song	Clear Line		
Menu Song	Set Bookmark	F8	Set a bookmark in the current song line
Menu Song	Clear Bookmark	Ctrl+F8	Clear the bookmark in the current song line
Menu Song	Delete current line	Ctrl+U	
Menu Song	Insert new empty line	Ctrl+I	
Menu Song	Insert new line with unused empty tracks	Ctrl+J	
Menu Song	Insert copy or clone of song line(s)...	Ctrl+K	
Menu Song	Insert new empty unused track to current song position	Ctrl+T	
Menu Song	Make a track duplicate to current song position	Ctrl+D	
Menu Song	Switch song between 4 or 8 channels...		
Menu Song	Song columns' order change/copy/clear...		
Menu Song	Change maximal length of tracks...		
Menu Song	All size optimizations...		
Menu Instrument	Copy		
Menu Instrument	Paste		
Menu Instrument / Paste special	Volume envelopes only		

Access Path	Entry	Accelerator Key	Action
Menu Instrument / Paste special	Envelope parameters only		
Menu Instrument / Paste special	Volume envelopes and Envelope parameters only		
Menu Instrument / Paste special	Insert Volume envelopes and Envelope parameters to cursor position		
Menu Instrument / Paste special	Volume L envelope only		
Menu Instrument / Paste special	Volume R envelope only		
Menu Instrument / Paste special	Volume R to L envelope only		
Menu Instrument / Paste special	Volume L to R envelope only		
Menu Instrument / Paste special	Table only		
Menu Instrument	Cut		
Menu Instrument	Delete		
Menu Instrument	Info about current instrument...		
Menu Instrument	Change all the instrument occurrences...		
Menu Instrument	Renumber all instruments...		
Menu Instrument	Load instrument from file...		
Menu Instrument	Save instrument as...		
Menu Instrument	Clear all unused instruments...		
Menu Instrument	All instruments cleanup...		
Menu Track	Copy		
Menu Track	Paste		
Menu Track	Cut		
Menu Track	Delete		

Access Path	Entry	Accelerator Key	Action
Menu Track	Increase Step Size	Num +	Increase the pattern step size
Menu Track	Decrease Step Size	Num -	Decrease the pattern step size
Menu Track	Info about current track...		
Menu Track	Search and build wise loop		
Menu Track	Expand loop		
Menu Track	Search and rebuild wise loops in all tracks...		
Menu Track	Expand loops in all tracks		
Menu Track	Renumber all tracks...		
Menu Track	Load track from file...		
Menu Track	Save track as...		
Menu Track	Clear all duplicated tracks, adjust song...		
Menu Track	Clear all tracks unused in song...		
Menu Track	All tracks cleanup...		
Menu Block / Tool Bar Block	Restore From Backup	Ctrl+B	Restore block from backup
Menu Block	Copy	Ctrl+C	
Menu Block	Paste	Ctrl+V	
Menu Block / Paste Special	Merge With Current Content	Ctrl+M	
Menu Block / Paste Special	Paste Only Volume Values		
Menu Block / Paste Special	Paste Only Speed Values		
Menu Block	Cut	Ctrl+X	
Menu Block	Delete	Del	
Menu Block	Exchange Block and Clipboard	Ctrl+E	
Menu Block / Tool Bar Block	Effects/Tools...	Ctrl+F	Effects/tools
Menu Block	Select All	Ctrl+A	
Menu Pokey	Pokey Chip Registers		
Menu Pokey	Activate Pokey Explorer Mode	Ctrl+Shift+F5	Activate POKEY Explorer Mode
Menu Pokey / AUDCTL	Bit 0 - Change Main Base Clock From 64 KHz To 15 KHz	C	

Access Path	Entry	Accelerator Key	Action
Menu Pokey / AUDCTL	Bit 1 - High Pass Filter Into Channel 2, Clocked By Channel 4	G	
Menu Pokey / AUDCTL	Bit 2 - High Pass Filter Into Channel 1, Clocked By Channel 3	F	
Menu Pokey / AUDCTL	Bit 3 - Join Channels 3 and 4 (16-bit Frequency)	K	
Menu Pokey / AUDCTL	Bit 4 - Join Channels 1 and 2 (16-Bit Frequency)	J	
Menu Pokey / AUDCTL	Bit 5 - Clock Channel 3 With 1.79 MHz	D	
Menu Pokey / AUDCTL	Bit 6 - Clock channel 1 with 1.79 MHz	A	
Menu Pokey / AUDCTL	Bit 7 - Change The 17-Bit Poly To 9-Bit Poly (Only For Distortion 0 and 8)	P	
Menu Pokey / SKCTL	Two Tone Mode	M	
Menu Pokey / Debug Channel	Next Channel	Enter	
Menu Pokey / Debug Channel	Previous Channel	Backspace	
Menu Pokey / Divisor	Increase By 0.1	+	
Menu Pokey / Divisor	Increase By 1.0	Shift++	
Menu Pokey / Divisor	Decrease By 0.1	-	
Menu Pokey / Divisor	Decrease By 1.0	Shift+-	
Menu Tools	Open ASMA		Open the Atari Sound Music Archive Website
Menu Tools	Open ASAP File...		Open an .SAP or other Atari Sound Module File
Menu Tools	Run Script...		Run an RMT script file (open, export, save) on the current session
Menu Tools	Options...		Display the configuration menu
Menu Tools	Tuning...		Display the tuning menu.
Menu Help	Help	F1	Open the local help in a browser.

Access Path	Entry	Accelerator Key	Action
Menu Help	Online Help	Shift+F1	Open the online help in a browser window
Menu Help	About RASTER Music Tracker		Display information about the app and its version in a pop-up dialog
Tool Bar Main	About		Display program information, version number and copyright
Tool Bar Main	MIDI on/off		MIDI on/off
Tool Bar Block	Transpose up		Note transposition up
Tool Bar Block	Transpose down		Note transposition down
Tool Bar Block	Change instrument		Instrument number change
Tool Bar Block	Change instrument		Instrument number change
Tool Bar Block	Volume up		Volume up
Tool Bar Block	Volume down		Volume down
Tool Bar Block	Block mode		Block modification mode
Tool Bar Block	Play block	Shift+F6	Play selected block
	Toggle NTSC/PAL	Ctrl+F12	Toggle NTSC/PAL

A.1 Note Keys

The "tonekeys" of the hotkey tables below: two rows of the keyboard form a piano, the lower row from C-1 and the upper row from C-2, with the row above each as the black keys. The layout follows the "Keyboard layout" option (QWERTY, AZERTY or the German QWERTZ; a first start picks it from the keyboard language). The pictures and tables are generated from the program (the `dump notekeys` script command): each keyboard row as its keys print them, with the note every key plays underneath, then the same as a table by note.

A.1.1 QWERTY

1	2	3	4	5	6	7	8	9	0	-	=
	C#2	D#2		F#2	G#2	A#2		C#3	D#3		F#3
Q	W	E	R	T	Y	U	I	O	P	[	]
C-2	D-2	E-2	F-2	G-2	A-2	B-2	C-3	D-3	E-3	F-3	G-3
A	S	D	F	G	H	J	K	L	;	'	
	C#1	D#1		F#1	G#1	A#1		C#2	D#2		
Z	X	C	V	B	N	M	,	.	/		

C-1	D-1	E-1	F-1	G-1	A-1	B-1	C-2	D-2	E-2
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

Note	Keys
------	------

C-1	Z
C#1	S
D-1	X
D#1	D
E-1	C
F-1	V
F#1	G
G-1	B
G#1	H
A-1	N
A#1	J
B-1	M
C-2	Q, ,
C#2	2, L
D-2	W, .
D#2	3, ;
E-2	E, /
F-2	R
F#2	5
G-2	T
G#2	6
A-2	Y
A#2	7
B-2	U
C-3	I
C#3	9
D-3	O
D#3	Ø
E-3	P
F-3	[
F#3	=
G-3	]

A.1.2 QWERTZ

1	2	3	4	5	6	7	8	9	0	ß	'
	C#2	D#2		F#2	G#2	A#2		C#3	D#3		F#3
	Q	W	E	R	T	Z	U	I	O	P	ü +
	C-2	D-2	E-2	F-2	G-2	A-2	B-2	C-3	D-3	E-3	F-3 G-3
	A	S	D	F	G	H	J	K	L	ö	ä #
	C#1	D#1		F#1	G#1	A#1		C#2	D#2		
	<	Y	X	C	V	B	N	M	,	.	-

C-1	D-1	E-1	F-1	G-1	A-1	B-1	C-2	D-2	E-2
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

Note	Keys
------	------

C-1	Y
C#1	S
D-1	X
D#1	D
E-1	C
F-1	V
F#1	G
G-1	B
G#1	H
A-1	N
A#1	J
B-1	M
C-2	Q, ,
C#2	2, L
D-2	W, .
D#2	3, ö
E-2	E, -
F-2	R
F#2	5
G-2	T
G#2	6
A-2	Z
A#2	7
B-2	U
C-3	I
C#3	9
D-3	0
D#3	ø
E-3	P
F-3	ü
F#3	'
G-3	+

A.1.3 AZERTY

&	é	"	'	(	-	è	_	ç	à	)	=
C#2	D#2			F#2	G#2	A#2		C#3	D#3		F#3
A	Z	E	R	T	Y	U	I	O	P	^	\$
C-2	D-2	E-2	F-2	G-2	A-2	B-2	C-3	D-3	E-3	F-3	G-3
Q	S	D	F	G	H	J	K	L	M	ù	*
C#1	D#1			F#1	G#1	A#1		C#2	D#2		
<	W	X	C	V	B	N	,	;	:	!	

C-1	D-1	E-1	F-1	G-1	A-1	B-1	C-2	D-2	E-2
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

Note	Keys
------	------

C-1	W
C#1	S
D-1	X
D#1	D
E-1	C
F-1	V
F#1	G
G-1	B
G#1	H
A-1	N
A#1	J
B-1	,
C-2	A, ;
C#2	é, L
D-2	Z, :
D#2	", M
E-2	E, !
F-2	R
F#2	(
G-2	T
G#2	-
A-2	Y
A#2	è
B-2	U
C-3	I
C#3	ç
D-3	O
D#3	à
E-3	P
F-3	^
F#3	=
G-3	\$

A.2 For All Edit Modes

A.2.1 Hotkeys

Key	Action
ESC	Stop playing song, mute all sounds. With the option "ESC resets Atari sound" (on by default) the sound routines are reinitialised as well.
F2	Switch to TRACK EDIT.
F3	Switch to INSTRUMENT EDIT.
SHIFT+F4	Go to INFO EDIT.
F4	Go to SONG EDIT.
F5	Play song from start.
F6	Play song and loop current pattern.
SHIFT+F6	Play and loop selection block.
F7	Play song from current cursor position.
SHIFT+F7	Play song from bookmark position (if set).
F8	Set bookmark to current cursor position.
CONTROL+F8	Clear bookmark.
F9	Mute/Unmute current channel.
CONTROL+F9	Solo current channel/Unmute all channels.
SHIFT+F9	Mute/Unmute all channels.
F11	RESPECT VOLUME mode on/off.
F12	Turn on/off autofollow mode.
CONTROL+F12	Toggle PAL/NTSC region.
PAGE UP, PAGE DOWN	Go to previous/next SONG line. In TRACK EDIT they move the cursor by the highlight step within the pattern track instead; use CONTROL+PAGE UP, CONTROL+PAGE DOWN for the SONG line there.
SHIFT+PAGE UP, SHIFT+PAGE DOWN	Go to next/previous subsong.
numblock /	Decrease octave for newly entered notes.
numblock *	Increase octave for newly entered notes.
numblock +, numblock -	Change pattern step size up/down.
CONTROL+numblock +, CONTROL+numblock -, SHIFT+numblock +, SHIFT+numblock -	Increase/decrease volume for newly entered notes. The mouse wheel over the VOLUME field of the song info does the same, the wheel over the OCTAVE field changes the octave.
SHIFT+tonekeys	Play note with current instrument and volume on currently active channel.
SHIFT+CONTROL+tonekeys	Play note with current instrument and volume on currently active channel in both stereo channels at once.
CONTROL+SPACE	Toggle between EDIT MODE and PROVE (JAM) MODE.
SHIFT	SHIFT is used during text input for either the SONG NAME or INSTRUMENT NAME, for uppercase and special characters.

Key	Action
CAPSLOCK	Toggle CAPSLOCK (indicated by "CAP" in the statusbar). CAPSLOCK also inverts the SHIFT key if it is held at the same time.
CONTROL+1-8	Turn on/off the channel 1 to 8.
CONTROL+O	Open RMT module.
CONTROL+R	Reload RMT module.
CONTROL+S	Save RMT module.
CONTROL+SHIFT+S	Save RMT module under a new name.
ALT+ENTER	Open the song properties.
CONTROL+N	Create new RMT module.
CONTROL+P	Print.
CONTROL+Y	Redo last change.
CONTROL+Z	Undo last change.
SHIFT+LEFT, SHIFT+RIGHT	Change active instrument.
Media PLAY	Play song from start/Stop.
Media NEXT, Media PREVIOUS	Play song from current pattern and skip next/previous SONG line.
ALT+F4	Exit RMT.

A.3 TRACK EDIT

A.3.1 Fields

NNN TT vV FSS

Field	Meaning
NNN	Notes, from C-1 to C-6.
TT	Instrument number, from \$00 to \$3F. If no note exists in highlighted row, the instrument column will behave like a note column instead.
vV	Volume, from \$0 to \$F Could also be used without a note or an instrument. "v" is simply added to clearly distinguish the volume value between the other columns.
FSS	Speed, from \$01 to \$FF. Fxx is intended to be a speed command, this is currently placeholder strings.

A.3.2 Hotkeys

Key	Action
UP, DOWN, LEFT, RIGHT, TAB, CTRL+TAB, SHIFT+TAB, PAGE UP, PAGE DOWN	Move cursor.
CONTROL+UP, CONTROL+DOWN	Go to next/previous SONG line.

Key	Action
CONTROL+PAGE UP, CONTROL+PAGE DOWN	Go to next/previous SONG line (the same as CONTROL+UP, CONTROL+DOWN).
CONTROL+LEFT, CONTROL+RIGHT	Change track pattern number at current position.
HOME	Move cursor to the start of current pattern track, or the start of a "wise loop".
CONTROL+HOME	Set/clear the start position of a "wise loop".
END	Move cursor to the end of current pattern track, or the end of a "wise loop".
CONTROL+END	Set/clear the end line of a pattern track.
SHIFT+CONTROL+ENTER	Set/clear the end line of a pattern track (the same as CONTROL+END).
ENTER	Play note at the cursor position. Every new ENTER hotkeys will also loop inside a selection block, if it exists.
SHIFT+ENTER	Play note at the cursor position and fetch its instrument and volume values, then become the active instrument.
CONTROL+ENTER	Play all notes at the currently edited line. (The option "Swap ENTER and CONTROL+ENTER" exchanges the two.)
CONTROL+D	Duplicate the current pattern track, and put it at the same place. (Note: If the pattern track is only used once in song, a messagebox asking for confirmation will appear.)
CONTROL+I, CONTROL+U	Insert/delete lines in the current pattern track.
INSERT, DELETE	Insert/delete lines in the current pattern track (the same as CONTROL+I, CONTROL+U). If a selection block exists, DELETE deletes its data instead.
CONTROL+G	Set "go to line" command in the song at current position.
CONTROL+T	Put new empty unused track to current song position and active channel.
SPACE	Delete note, instrument, volume and speed values in the track at the cursor position.

A.3.3 Note Column

Key	Action
tonekeys	Insert note and play it. (Note: If RESPECT VOLUME is active, previous volume value won't be replaced.)
numblock 1-6	Change octave of the note at the cursor position and play it.

Key	Action
BACKSPACE	Delete note and instrument values (volume and speed values will not be cleared).

A.3.4 Instrument Number Column

Key	Action
tonekeys	Insert note and play it. (Note: This will only work if the note column row is empty.)
0 - F	Set up/change instrument number.
BACKSPACE	Delete note and instrument values (volume and speed values will not be cleared).

A.3.5 Volume Column

Key	Action
0 - F	Set up/change volume value.
BACKSPACE	Delete note, instrument and volume values (speed values will not be cleared).

A.3.6 Speed Column

Key	Action
0 - F	Set up/change speed value.
BACKSPACE	Delete speed values.

A.3.7 Block Operations

Key	Action
SHIFT+UP, SHIFT+DOWN, SHIFT+HOME, SHIFT+END	Select block data.
ESCAPE	Deselect block data.
DELETE	Delete selection block data.
HOME, END	Go to the start/end of the selection block.
CONTROL+A	Select all valid data from the currently edited pattern track, and form a selection block.
CONTROL+B	Restore selection block data from backup. The backup is created as soon as the selection block is being formed.
CONTROL+C	Copy selection block to the clipboard. If no selection exists, the data at cursor position will be taken.

Key	Action
CONTROL+E	Exchange of selection block data and clipboard data.
CONTROL+M	Paste and merge data from the clipboard to the cursor position.
CONTROL+V	Paste data from the clipboard to the cursor position.
CONTROL+X	Cut selection block to the clipboard. If no selection exists, the data at cursor position will be taken.

A.3.8 Block Data Modifications

Key	Action
SHIFT+CONTROL+A	Block modifications mode. All changes are provided either on each line of the block, or only on lines with instrument number equal to active instrument.
CONTROL+F2, CONTROL+F1	Transpose note up / down by semitones (5 octaves max). If no selection exists, the data at cursor position will be taken.
CONTROL+F4, CONTROL+F3	Transpose note up / down by octaves (5 octaves max). If no selection exists, the data at cursor position will be taken.
SHIFT+CONTROL+LEFT, SHIFT+CONTROL+RIGHT	Change instrument values, from \$00 to \$3F max. If no selection exists, the data at cursor position will be taken.
SHIFT+CONTROL+UP, SHIFT+CONTROL+DOWN	Volume up/down, from \$0 to \$F max. If no selection exists, the data at cursor position will be taken.
CONTROL+F	Display the Block effects/tools window.

A.3.9 Mouse control

A.3.10 Tracks top area (Shown as TRACK L1 to TRACK R4 above pattern tracks)

Key	Action
LeftMouseButton	Turn channel on/off.
RightMouseButton	Solo channel/turn back on all channels.

A.4 INSTRUMENT EDIT

A.4.1 Fields

Field	Meaning
NAME	Name of the instrument, 32 chars max.
ENVELOPE LENGTH	Length of the envelope, from \$01 to \$20 (i.e. 32 max.).
ENVELOPE GOTO	Jump to given envelope position when the envelope end is reached.
FADEOUT	Volume slide when the end of envelope is reached for the first time. \$00 = no volume slide, \$FF = maximal volume slide.
VOL MIN	Minimal volume value reached by FADEOUT parameter.
TABLE LENGTH	Length of the table, from \$01 to \$20 steps (i.e. 32 max.).
TABLE GOTO	Jump to given table position when the table end is reached.
TABLE SPEED	Speed for the each table step, from \$01 to \$40 vbi.
TABLE TYPE	Table type. 0 = notes, 1 = frequencies
TABLE MODE	Table mode. 0 (SET) = add note (or frequency) to base note, 1 (ADD) = add note (or frequency) to the last calculated note (or frequency). Note: If the resulting note is outside of the C-1 to C-6 range (hex values \$00 to \$3D), the output volume will be zero. Frequency additions are not limited however.
EFFECT DELAY	Delay before the start of VIBRATO and FREQSHIFT effects by \$01 to \$FF vbi, \$00 = no effects.
EFFECT VIBRATO	Vibrato effect, 3 preset levels from \$01 to \$03, \$00 = no vibrato.
EFFECT FREQSHIFT	Frequency shifting effect, from \$00 to \$FF for each vbi.
AUDCTL 15KHZ	Turn on/off AUDCTL bit 0, "Change main base clock from 64 KHz to 15 KHz".
AUDCTL HPF 2+4	Turn on/off AUDCTL bit 1, "High pass filter into channel 2, clocked by channel 4".
AUDCTL HPF 1+3	Turn on/off AUDCTL bit 2, "High pass filter into channel 1, clocked by channel 3".
AUDCTL JOIN 3+4	Turn on/off AUDCTL bit 3, "Join channels 3 and 4 (16-bit frequency)".
AUDCTL JOIN 1+2	Turn on/off AUDCTL bit 4, "Join channels 1 and 2 (16-bit frequency)".
AUDCTL 1.79 CH3	Turn on/off AUDCTL bit 5, "Clock channel 3 with 1.79 MHz".
AUDCTL 1.79 CH1	Turn on/off AUDCTL bit 6, "Clock channel 1 with 1.79 MHz".
AUDCTL POLY9	Turn on/off AUDCTL bit 7, "Change the 17-bit poly to 9-bit poly (Distortion 0 and 8 only)".

A.4.2 Table of Notes/Frequencies

Field	Meaning
TABLE \$00-\$1F	From 1 to 32 steps per table. Values can be either notes (in semitones), or frequencies, based on the TABLE TYPE value. Values can range from positive (\$00 to 7F) or negative (FF to \$80) numbers.

A.4.3 Envelope

Field	Meaning
VOLUME R	Volume values from \$0 to \$F, used only for stereo songs in tracks R1 to R4.
VOLUME L	Volume values from \$0 to \$F used in tracks L1 to L4.
DISTORTION	POKEY AUDC Distortion values, from \$0 to \$E. Only even values can be used, otherwise Volume Only output would occur.
COMMAND	Envelope command, from \$0 to \$7.
X, Y	Both parameters make use of values from \$0 to \$F for effect commands. They can be used as a two separate parameters, or as one 8-bit hexadecimal value \$XY.
AUTOFILTER	Automatic High Pass Filter envelope effect. \$0 = not active, \$1 = active. It will only work with channels 1 or 2. This effect automatically enables the AUDCTL High Pass Filter bits for each channel, where HPF 1+3 or HPF 2+4 may be output. It has higher priority over non filtered voices, and will hijack them while the required conditions are met. If the resulting volume is zero, filtering won't be enabled, and won't take priority over other channels. Channels 3 and 4 won't be muted, so the resulting sound could be modulated into a richer tone if desired.
PORTAMENTO	Portamento. \$0 = not active, \$1 = active. If active, the "Portamento volatile frequency" is used instead of the current frequency. Portamento is mostly useful for producing pitch bending effects.

A.4.4 Sound Type (DISTORTION Parameter)

Value	Meaning
0	Distortion 0, white noise. (AUDC \$0v, Poly5+17/9)
2	Distortion 2, square-ish tones. (AUDC \$2v, Poly5)
4	Distortion 4, no note table yet, Pure Table by default. (AUDC \$4v, Poly4+5)
6	Distortion C, fallback setting, buzzy bass tones. (AUDC \$Cv, Poly4)
8	Distortion 8, white noise. (AUDC \$8v, Poly17/9)
A	Distortion A, pure tones. (AUDC \$Av)
C	Distortion C, buzzy bass tones. (AUDC \$Cv, Poly4)
E	Distortion C, gritty bass tones. (AUDC \$Cv, Poly4)

A.4.5 Envelope command (COMMAND parameter)

Value	Meaning
0	Play the base note shifted by \$XY semitones. If the resulting note is outside the C-1 to C-6 range (hex values \$00 to \$3D), the volume output will be zero.
1	Play the frequency \$XY directly.
2	Play the base note shifted by frequency \$XY.
3	Add \$XY semitones to base note. Play base note (new value). If the resulting note is outside the C-1 to C-6 range (hex values \$00 to \$3D), the volume output will be zero.
4	Add frequency \$XY to FSHIFT register. Play base note.
5	Set up portamento speed \$X, step \$Y. Each \$X vbi will be "volatile portamento frequency" shifted up or down by the \$Y value. If \$XY=\$00, the current frequency will be used directly as volatile portamento frequency.
6	Add \$XY value to FILTER_SHFRQ. Whenever a new note in track is playing, FILTER_SHFRQ is initialized to \$01, so by default the Automatic Filter channels frequencies are offset by 1 unit in order to produce SID-like sound.
7	Set the instrument AUDCTL directly using \$XY values. Play BASE_NOTE. Exceptions: \$FF = VOLUME ONLY mode. FE/FD = enable/disable Two-Tone Filter (only for channel 1, modulated by channel 2).

A.4.6 Hotkeys

Key	Action
LEFT, RIGHT, UP, DOWN	Move cursor.
TAB	Move cursor to the instrument parameters, the envelope parameters or the table parameters in succession
SHIFT+TAB	Move cursor to the instrument name line.
0-F, CONTROL+LEFT, CONTROL+UP, CONTROL+RIGHT, CONTROL+DOWN, BACKSPACE, SPACE, DELETE	Change parameter values (in valid ranges only).
HOME	In the parameters: move cursor to the first parameter (ENVELOPE LENGTH).
SHIFT+CONTROL+UP, SHIFT+CONTROL+DOWN	Change values for all steps of instrument envelope/table at cursor position.
SHIFT+CONTROL+numblock +, SHIFT+CONTROL+numblock -	Change the L+R volume envelopes up/down. If the cursor is on the "VOLUME L" or "VOLUME R" line, only the volume envelope for either line will be changed.

A.4.7 Table of notes/frequencies

Key	Action
HOME	Move cursor to the start of the table/to the start of "table loop".
CONTROL+HOME	Set up the start of "table loop".
END	Move cursor to the end of the table.
CONTROL+END	Set up the end of the table.
INSERT	Insert empty step into the table at cursor position and shift to the right.
SHIFT+INSERT	Duplicate step at cursor position and shift to the right.
DELETE	Delete step at cursor position and shift to the left.
SPACE	Clear the table step at cursor position and move cursor to the right.

A.4.8 Envelope Parameters

Key	Action
HOME	Move cursor to the start of the envelope/to the start of "envelope loop".
CONTROL+HOME	Set up the start of "envelope loop".
END	Move cursor to the end of the envelope.
CONTROL+END	Set up the end of the envelope/set up a maximal length of the envelope.
INSERT	Insert empty step into the envelope at cursor position and shift to the right.
SHIFT+INSERT	Duplicate step at cursor position and shift to the right.
DELETE	Delete step at cursor position and shift to the left.
SPACE	Clear the envelope step at cursor position and move cursor to the right.

A.4.9 Mouse control

A.4.10 Volume Envelope Area

Key	Action
LeftMouseButton	Draw the volume envelope.
RightMouseButton	Set the volume to zero in the envelope.

A.5 INFO EDIT

Field	Meaning
NAME	Name of the song and author, 64 chars max.

MUSIC SPEED: AA/MM/S

Field	Meaning
AA	Current song speed, from \$01 to \$FF.
MM	Main speed, from \$01 to \$FF. By default the main speed is used when a song plays from the beginning.
S	Engine speed, from \$1 to \$8. (X player calls per frame.) Technically, speed above 4 is unsupported in the exported formats.

A.5.1 Hotkeys

Key	Action
LEFT, RIGHT, TAB	Move cursor.
SHIFT+TAB	Move cursor song name line.
0-F, CONTROL+LEFT, CONTROL+UP, CONTROL+RIGHT, CONTROL+DOWN	Change parameter values (in valid ranges only).
ENTER	Return from INFO EDIT to the currently active screen (Tracks or Instruments edit).

A.5.2 Mouse control

A.5.3 MAXTRACKLENGTH, MONO/STEREO, PAL/NTSC

Key	Action
LeftMouseButton	Toggle or prompt dialog boxes for each elements, then apply changes to the song.

A.6 SONG EDIT

Key	Action
LEFT, RIGHT, UP, DOWN, PAGE UP, PAGE DOWN, TAB, SHIFT+TAB	Move cursor.
HOME	Move cursor to the start of the song.
END	Move cursor to last song line with valid data.
SHIFT+PAGE UP	Move cursor to the start of current subsong or start of previous subsong if the action is repeated.
SHIFT+PAGE DOWN	Move cursor to start of next subsong.
0-F, CONTROL+LEFT, CONTROL+RIGHT, BACKSPACE	Change the pattern track number or "go to line" value in the song.

Key	Action
INSERT, CONTROL+I, DELETE, CONTROL+U	Insert/delete lines in the song (with auto-change of all relevant "go to line" values).
CONTROL+D	Duplicate the current pattern track, and put it at the same place. (Note: If the pattern track is only used once in song, a messagebox asking for confirmation will appear.)
CONTROL+G	Set "go to line" command in the song at current position.
CONTROL+T	Put new empty unused track to current song position and active channel.
CONTROL+K	Insert copy or clone of song line(s).
CONTROL+J	Prepare song line with unused empty tracks.
ENTER	Return from SONG EDIT to the currently active screen (Tracks or Instruments edit).

A.7 PROVE (JAM) MODE

Navigation is identical in TRACKS and SONG areas, but editing is disabled, and replaced with tonekeys always playing notes. INSTRUMENTS and INFO areas will still behave the same as they are in EDIT MODE, simply hold SHIFT to test notes, except when the text is being edited.

APPENDIX B

The RMT module file format

The format of the .rmt files, for the programmers who read them. It is the format version 1, used by all the RMT trackers up to now; the description is the official one of the repository.

This is the module file format used by RMT trackers with version 1.xy.

B.0.1 Segment Binary Load Header (Mandatory)

RMT module files are regular multi-segment binary load files. They start with a 6-byte binary load header. It includes the start address and end address in a virtual 64k address space. The absolute addresses in the segments and structures can be relocated based on their distance from the start address.

> Offset	Type	Description
-----	----	-----
\$00	WORD	Magic byte sequence
\$FFFF	\$02 WORD	Start address, default is \$4000
	\$04 WORD	End address

B.0.2 Module Header Struct

Offset	Type	Description
\$00 WORD	Header	er string "RMT4" for mono or "RMT8" for stereo modules
\$04 BYTE	Track	k length (\$00 means 256)
\$05 BYTE	Song	speed
\$06 BYTE	Repl	ay frequency
\$07 BYTE	Form	at version number (\$01 for player routine 1.x compatible format \$02 for 2.x)
\$08 WORD	Addr	ess of the instruments table
\$0a WORD	Addr	ess of the low-byte table of the track addresses
\$0c WORD	Addr	ess of the high-byte table of the track addresses
\$0e WORD	Addr	ess of the track sequence table

B.0.3 Instrument Struct

Offset	Type	Description
\$00 BYTE	tlen	(pointer to the end of the table of notes)
\$01 BYTE	tgo	(pointer to the loop of the table of notes)
\$02 BYTE	elen	(pointer to the end of the envelope)
\$03 BYTE	ego	(pointer to the loop of the envelope)
\$04 BYTE	tspd	(bit 0-5), tmode (bit 6), ttype (bit 7)

Offset	Type	Description
\$05 BYTE	AUDC	TL
\$06 BYTE	Vsli	de
\$07 BYTE	Vmin	(bit 4-7)
\$08 BYTE	Dela	y (\$00 for no vibrato & no fshift)
09 BYTE	Vibr	ato
0a BYTE	fshi	ft
0b BYTE	Unus	ed
0c	Tabl	e of notes
??	Enve	lope

B.0.4 Table of Notes Struct

Type	Description
BYTE	Note or frequency (according to the ttype)

B.0.5 Envelope Struct

Type	Description
BYTE	Volume (bits 0-3 for the left channel, bits 4-7 for the right channel (in RMT4 it's the same as bits 0-3))
BYTE	Portamento (bit 0), distortion (bit 1-3), command (bit 4-6), filter (bit 7)
BYTE	XY

B.0.6 Track Struct

Type	Description
BYTE	
bit 0-5	note
bit 6-7	volume (HI) or pause (1-3 beats) or special

```
if note is $00-$3c:
BYTE
    bit 0-1 volume (L0)
    bit 2-7 instrument number

if note is $3d:
BYTE
    bit 0-1 volume (L0) volume only

if note is $3e:
    bit 6-7 pause
    if pause is $01-$03:    pause 1-3 beats
    if pause is $00:      next byte pause 1-255 beats

if note is $3f:
    if bit 6-7 is zero: next byte speed $01-$ff
    if bit 6 is zero, 7 is set up: next byte is track jump pointer (go to $00-$ff from the
    begin of track data)
    if bit 6-7 is set up:  END of track
```

B.0.7 Instruments Table

Type	Description
WORD	ptr_instr0
WORD	ptr_instr1
WORD	ptr_instr3
...	

B.0.8 Track Addresses Table

Type	Description
BYTE	lowbyte_of_ptr_track0
BYTE	lowbyte_of_ptr_track1
BYTE	lowbyte_of_ptr_track2
...	

```
BYTE    highbyte_of_ptr_track0
BYTE    highbyte_of_ptr_track1
BYTE    highbyte_of_ptr_track2
...
```

B.0.9 Track Sequence Table

Type	Description
BYTE	tracknumL1,tracknumL2,tracknumL3,tracknumL4,[tracknumR1,...,tracknumR4]
BYTE	tracknumL1,tracknumL2,tracknumL3,tracknumL4,[tracknumR1,...,tracknumR4]
BYTE	tracknumL1,tracknumL2,tracknumL3,tracknumL4,[tracknumR1,...,tracknumR4]
...	

if tracknum is FF, then an empty track is used

if tracknumL1 is FE, then gotoline(BYTE)=tracknumL2, goto_pointer(WORD)=(tracknumL3,4)

Note: gotoline(BYTE) is not used in the player (but the tracker uses it)

APPENDIX C

MIDI implementation chart

The MIDI implementation chart of the multitimbral instrument that the tracker is, from the original documentation of **RMT** (the program also accepts the controllers described in chapter 13).

```
*****
RASTER MUSIC TRACKER - RMT
by Radek Sterba, Raster/C.P.U., 2002-2008
http://raster.atari.org
*****
```

RMT [Raster Music Tracker] MULTITIMBRAL INSTRUMENT - MIDI IMPLEMENTATION CHART

Function		Recognized	Remarks
Basic Channel	Default	2-9	2-5 = L1-L4 , 6-9 = R1-R4
	Changed	x	(Channel 1 is reserved for edit)
Mode	Default	Mode 4	
	Messages	x	
	Altered	x	
Note Number :	True voice	36-96	RMT notes from C-1 to C-6
		36-96	
Velocity	Note ON	o 9nH,v=1-127	RMT volume = v / 8
	Note OFF	x 9nH,v=0 8nH,v=xx	
After Touch	Key's	x	
	Ch's	x	
Pitch Bend		x	
Control Change		x	
Prog Change : True #		x	
System exclusive		o 0-63	RMT instruments 00-3F
		o 0-63	
Common : Song Pos.		x	
Song Sel.		x	
Tune		x	
System : Clock		x	
Real Time: Commands		x	
Aux : LocalON/OFF		x	
Messages :	ResetAllCtr	o (121)	Silence + Reinit of RMT routine
	AllNotesOFF	o (123)	Note OFF
	ActiveSense	x	
	SystemReset	o	Silence + Reinit of RMT routine
Mode 1: OMNI ON, POLY		Mode 2: OMNI ON, MONO	o: Yes
Mode 3: OMNI OFF, POLY		Mode 4: OMNI OFF, MONO	x: No

APPENDIX D

Building from the source

RITMO is built with CMake and Qt. The full instructions are in the file `BUILD.md` of the source; here is the short version.

D.1 Linux

```
sudo apt install cmake qt6-base-dev portaudio19-dev librtmidi-dev
cmake -B build-qt -DCMAKE_BUILD_TYPE=Release
cmake --build build-qt -j
./build-qt/out/ritmo song.rmt
```

The program is `build-qt/out/ritmo`, with the `resources` folder next to it. `-DRMT_QT_MAJOR=5` builds with Qt 5.15 instead of Qt 6.

D.2 Windows and macOS

Windows is built with MSYS2 (MinGW64: GCC, Qt 6, PortAudio, RtMidi) and macOS with the Apple compiler and the Qt of the official installer. The GitHub workflows of the repository (`.github/workflows`) build the packages of both, and the AppImage for Linux (`scripts/build-appimage.sh`), and are the best description of the steps.

D.3 Core only and tests

`-DRMT_BUILD_CORE_ONLY=ON` builds the engine and the audio and MIDI backends without Qt, and with `-DRMT_CORE_TEST=ON` the `RmtCoreTest` program, a smoke test of the engine. The `scripts/test-scripting.sh` script runs the reference scripts.

D.4 Test hooks

For tests and documentation the program can run without a display (`QT_QPA_PLATFORM=offscreen`). The environment variables `RMT_QT_GRAB` (save the window and quit), `RMT_QT_KEYS`, `RMT_QT_COMMANDS`, `RMT_QT_FILEDIALOG`, `RMT_QT_DIALOG` and `RMT_QT_MENU_GRAB` (the menus as images) press keys, run menu commands, answer the dialogs and save them. The screenshots of this manual were taken that way, by `doc/manual/make-screenshots.sh`.

Credits and licenses

E.1 Credits

- Radek Štěřba, Raster/C.P.U., 2002–2009 (R.I.P.). The original RASTER Music Tracker. Thank you for everything you did; we truly miss you.
- Robert Petruzela (Bob!k/C.P.U.) and JirkaS/C.P.U.
- Vin Samuel (VinsCool), 2021–2024: RMT 1.31 to 1.34, the POKEY tuning calculations, the SAP-R dumper and the VU-Player, and much more.
- Peter Dell (JAC!), 2024 to present: RMT 1.35 and the base of this fork.
- Gianluca Renzi, 2026: the Qt port and the RITMO fork (Linux, Windows, macOS).

Also: the LZSS compression is by DMSC, ported to C++ by VinsCool; the unrolled LZSS music driver is by Rensoupp; the bitmap graphics, the ideas and the beta testing are by PG; advice on tuning from synthpopalooza and OPNA2608; ideas and suggestions from Enderdude, Spring, Ivop, Tatqoo and Miker. Thanks to Fox/Taquart (XASM and ASAP), Jaskier/Taquart (TMC and optimizations of the RMT routines), the people of the Chiptune Café, AtariAge and GBAtemp, the Polish Atarians of Atariarea and all the 8-bit Atarians of the world.

E.2 The fonts

The logo and the icon of **RITMO**, and the cover of this manual, are drawn with the glyphs of the IBM VGA 9x16 font, in the version *PxPlus IBM VGA 9x16* of *The Ultimate Oldschool PC Font Pack* by VileR (<https://int10h.org/oldschool-pc-fonts/>), licensed under CC BY-SA 4.0.

E.3 Disclaimer

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